

TEMPORAL LOCALIZATION OF SCORE ERROR AND THE ODDS FOR STOCHASTIC SAMPLING IN DIFFUSION MODELS

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ABSTRACT. Whether stochastic reverse-time sampling improves on the probability-flow ODE depends not only on how large the score error is, but on *where along the sampling trajectory* it is concentrated. We study this dependence in a one-dimensional Gaussian model in which the learned score is affine in space with arbitrary time-dependent coefficients. Written in the first-order score-error coordinates (a, u) and a logarithmic variance clock, the reverse moment equations are linear and exactly solvable, and the terminal Kullback–Leibler divergence is, to second order, a sum of squares of two linear functionals of the error profile. The expansion is controlled by an L^1 norm in the log-variance clock rather than by pointwise smallness, which makes it applicable to trained networks whose error is of order one at the lowest noise levels.

The two functionals are exponential filters. In physical units, stochasticity damps shape error above the noise level $\sigma^*(\gamma) = \sigma_0 \sqrt{(1 + \gamma)^{1/\gamma} - 1}$ and amplifies it below; for the standard reverse SDE ($\gamma = 1$) the crossover is exactly the data scale σ_0 . Motivated by score-error profiles measured on trained networks, we then analyze power-law profiles $(\sigma_0/\sigma)^p$. All functionals become (incomplete) Beta functions. For the shape mode there is a critical exponent $p_c(\gamma)$, characterized by a Gamma-function identity, with $p_c(1) = 1$ exactly and $p_c(0^+) \approx 1.18$; for the mean mode the critical exponent is 0 for every γ . The same formulas give the optimal constant stochasticity and the largest useful one.

We then quantify the odds that stochasticity helps, randomizing some parameter choices within empirically-motivated models. A scalar random amplitude multiplying a deterministic profile produces degenerate odds; two independent mode amplitudes give closed arctangent, arcsine or probit laws. When the profile is a Gaussian process with a power-law envelope, the odds reduce to bivariate Gaussian orthant probabilities, and the correlation length of the process interpolates between a sharp phase transition and an explicit Beta-function odds curve. A static-parameter benchmark, in which all error parameters are constant in time, is recovered as the zero-exponent member of the family. Finally, we confront the theory with 144 checkpoints trained on a one-dimensional Gaussian mixture.

Draft status (remove before circulation). This version fuses the drafts “*Temporal localization of score error and the odds for stochastic sampling*” (time-dependent profiles, finite γ) and “*The odds for stochastic sampling in diffusion models*” (static parameters, general priors, local $\gamma \downarrow 0$). Notation is unified as follows: horizon Λ (formerly $\lambda = -\log \zeta_T$), linear shape coordinate $a = 1 - 1/\alpha_\theta$ (formerly $u = 1/\alpha_\theta - 1$ in the static draft), mean coordinate u , distance to the data end d , relative noise $y = \sigma^2/\sigma_0^2$. Sections 5–7 and [Lemma 2.2](#) and [Proposition 4.1](#) are new; [Section 3](#), [Section 8](#) and the appendices are condensed from the static draft; [Section 9](#) contains provisional numbers from notebook 056 and TODO placeholders.

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1. INTRODUCTION

Sampling from a score-based diffusion model can be performed with the probability-flow ODE or with a family of reverse-time SDEs of varying stochasticity [1, 9, 3]. With exact score functions and exact prior distributions, all these samplers share the same marginals. With a learned score they do not, and neither deterministic nor stochastic sampling is preferable in principle. A KL-divergence analysis of the reverse dynamics [7] identifies the competing mechanisms:

stochasticity contracts discrepancies already accumulated in the generated density, but it also increases the weight of the score error currently encountered. The net effect should therefore depend on the *temporal localization* of score error, and not only on an integrated measure of its size.

The purpose of this paper is to provide a more mathematically complete analysis of this phenomenon in a setting where all quantities of interest can be computed. We consider the one-dimensional Gaussian setting of [7, Section 5], and generalize the misspecified score parameters to be arbitrary functions of time. Three observations make the resulting model tractable and, we believe, relevant in practical settings.

Linearity in the right coordinates. Writing the affine score error as $\epsilon_t(x) = \sigma_0 u(t)/V(t) + a(t)(x - \mu_0)/V(t)$, the reverse mean and variance obey linear equations in a logarithmic variance clock ℓ , whose exact solutions are integrating-factor formulas (Proposition 2.1). To first order, the terminal errors are two linear functionals $L_\gamma[a]$ and $M_\gamma[u]$, with a remainder controlled by $\|a\|_{L^1(d\ell)}$ (Lemma 2.2). Because the clock compresses the low-noise region, $d\ell \approx d\sigma^2/\sigma_0^2$ for $\sigma \ll \sigma_0$, a score error of order one at the lowest noise levels still produces a small terminal perturbation. On trained networks the first-order response is indistinguishable from the full Gaussian moment dynamics (Section 9).

Stochasticity as a filter, in physical units. Relative to the ODE, stochastic sampling reweights shape error at log-variance distance d from the data end by $(1 + \gamma)e^{-\gamma d}$. This weight exceeds one exactly below the noise level $\sigma^*(\gamma) = \sigma_0 \sqrt{(1 + \gamma)^{1/\gamma} - 1}$ (Proposition 4.1). Stochasticity therefore amplifies score error below σ^* and damps it above; for the usual reverse SDE, $\gamma = 1$, the crossover is the data scale itself.

Power laws. Score-error profiles of networks trained on a Gaussian mixture are well described, in root-mean-square magnitude across checkpoints, by piecewise power laws in the noise level σ with two or three exponents (Section 9). A power law $(\sigma_0/\sigma)^p$ has an integrable singularity at the data end, precisely where stochastic sampling gives error its largest weight. This produces a phase diagram with a γ -dependent critical shape exponent $p_c(\gamma)$, with $p_c(1) = 1$ exactly (Theorem 5.3), whereas the mean mode has the universal threshold $p_u = 0$ (Theorem 5.9). All the relevant quantities are Beta functions, including the optimal constant stochasticity (Proposition 5.5).

The second theme is probabilistic. Following [7], we regard a trained network as one draw from a population of imperfect scores and ask for the probability, or “odds”, that stochasticity improves the generated distribution. Near the exact score the KL comparison sees only a few linear projections of the error profile, so the odds reduce to orthant probabilities of low-dimensional Gaussian vectors. We treat two random models for the observed experimental data. First, deterministic power-law profiles with random amplitudes (Section 6); a single scalar amplitude gives degenerate odds, while independent mean and shape amplitudes give closed arctangent, arcsine or probit laws. Second, Gaussian processes with a power-law envelope (Section 7); here the correlation length of the process interpolates between a sharp phase transition and an explicit odds curve.

Finally, the static model of [7], in which all misspecification parameters are constant in time, is the zero-exponent member of this family. Its results for general positive priors are collected in Section 3 and Section A. At the horizons used in practice, $\Lambda = \log(V(T)/\sigma_0^2) \approx 10$, the static model predicts that stochasticity helps with probability close to one for every atomless prior (Proposition 3.5). Time-localized profiles explain why this need not be so.

Outline. Section 2 sets up the model and proves the exact linearization. Section 3 is the static benchmark. Section 4 develops the filter picture. Section 5 analyzes power-law profiles. Sections 6 and 7 treat the odds. Section 8 collects exact nonperturbative results. Section 9 compares with trained networks, and Section 10 discusses limitations and open problems.

2. TIME-DEPENDENT GAUSSIAN SCORE MODEL

2.1. Forward process, clocks and reverse samplers. Fix $T > 0$ and consider the one-dimensional variance-exploding forward diffusion

$$(2.1) \quad dX_t = g(t) dW_t, \quad X_0 \sim p_0 = \mathcal{N}(\mu_0, \sigma_0^2), \quad 0 \leq t \leq T.$$

Write $\sigma(t)^2 := \int_0^t g(s)^2 ds$ for the added noise variance (so $\sigma(t) = t$ for the EDM schedule [3]) and

$$(2.2) \quad V(t) := \sigma_0^2 + \sigma(t)^2, \quad p_t = \mathcal{N}(\mu_0, V(t)), \quad \nabla \log p_t(x) = -\frac{x - \mu_0}{V(t)}.$$

We use four equivalent parametrizations of time:

$$(2.3) \quad \Lambda := \log \frac{V(T)}{\sigma_0^2}, \quad \ell := \log \frac{V(T)}{V(t)} \in [0, \Lambda], \quad d := \Lambda - \ell = \log \frac{V(t)}{\sigma_0^2}, \quad y := \frac{\sigma(t)^2}{\sigma_0^2} = e^d - 1.$$

Here ℓ is the sampling clock (it increases during generation), d is the log-variance distance from the data end, and y is the added noise variance relative to the data variance. We also write $\zeta_T := e^{-\Lambda} = \sigma_0^2/V(T)$ as in [7]. For the EDM values $\sigma_0 = 0.5$, $\sigma(T) = 80$ one has $\Lambda \approx 10.15$; for the Gaussian mixture of Section 9, $\sigma_0^2 = 0.1219$ and $\Lambda \approx 10.87$.

Let $\tau \in [0, T]$ be reverse time, so the forward time is $t = T - \tau$ and $d\ell/d\tau = g(t)^2/V(t)$. For a constant stochasticity level $\gamma \geq 0$ consider

$$(2.4) \quad dY_\tau = \frac{1+\gamma}{2} g(t)^2 s_\theta(Y_\tau, t) d\tau + \sqrt{\gamma} g(t) dB_\tau, \quad t = T - \tau.$$

The case $\gamma = 0$ is the probability-flow ODE and $\gamma = 1$ the usual reverse SDE.

2.2. Affine scores and first-order error coordinates. We assume that the learned score is affine in space with arbitrary time-dependent coefficients,

$$(2.5) \quad s_\theta(x, t) = -\frac{x - \mu_\theta(t)}{\alpha_\theta(t)V(t)}, \quad \alpha_\theta(t) > 0,$$

and that the reverse process starts from $Y_0 \sim \mathcal{N}(\mu_T, \beta_T V(T))$. It is convenient to measure the error by the coefficients in which the score error is linear:

$$(2.6) \quad a(t) := 1 - \frac{1}{\alpha_\theta(t)}, \quad u(t) := \frac{\mu_\theta(t) - \mu_0}{\alpha_\theta(t)\sigma_0}, \quad r_T := \frac{\mu_T - \mu_0}{\sigma_0}, \quad b_T := \beta_T - 1.$$

Then

$$(2.7) \quad \epsilon_t(x) := s_\theta(x, t) - \nabla \log p_t(x) = \frac{\sigma_0 u(t)}{V(t)} + a(t) \frac{x - \mu_0}{V(t)}.$$

Conversely, every affine score $s(x, t) = -(x - \mu_0)/V(t) + b(t) + C(t)(x - \mu_0)$ has this form with $a = VC$ and $u = Vb/\sigma_0$; a positive α_θ exists if and only if $a < 1$. All results below that are stated in terms of (a, u) hold for arbitrary real a . This matters empirically: least-squares projections of a trained network's error onto affine functions produce (a, u) directly, and occasionally $a \geq 1$ (Section 9). The coordinates of [7] are related by

$$\log \alpha_\theta = -\log(1 - a) = a + O(a^2), \quad \frac{\mu_\theta - \mu_0}{\sigma_0} = \frac{u}{1 - a} = u + O(|a||u|).$$

Throughout, a and u are regarded as functions of the clock ℓ (or of d , y or σ), with a slight abuse of notation.

2.3. Exact moment dynamics. Let $r_\gamma(\ell) := (\mathbb{E}Y_\tau - \mu_0)/\sigma_0$ and $v_\gamma(\ell) := \text{Var}(Y_\tau)/V(t)$, where $\ell = \ell(\tau)$.

Proposition 2.1 (Exact moment dynamics). *For the reverse process (2.4)–(2.5), the law of Y_τ is Gaussian for every τ , and in the clock ℓ ,*

$$(2.8) \quad r'_\gamma(\ell) = -\frac{1+\gamma}{2} \left[(1-a(\ell))r_\gamma(\ell) - u(\ell) \right], \quad r_\gamma(0) = r_T,$$

$$(2.9) \quad v'_\gamma(\ell) = [-\gamma + (1+\gamma)a(\ell)]v_\gamma(\ell) + \gamma, \quad v_\gamma(0) = \beta_T.$$

Consequently, with $I_a(s) := \int_s^\Lambda a(r) \, dr$,

$$(2.10) \quad r_\gamma(\Lambda) = e^{-\frac{1+\gamma}{2}(\Lambda-I_a(0))} r_T + \frac{1+\gamma}{2} \int_0^\Lambda e^{-\frac{1+\gamma}{2}[(\Lambda-s)-I_a(s)]} u(s) \, ds,$$

$$(2.11) \quad v_\gamma(\Lambda) = e^{-\gamma\Lambda+(1+\gamma)I_a(0)} \beta_T + \gamma \int_0^\Lambda e^{-\gamma(\Lambda-s)+(1+\gamma)I_a(s)} \, ds,$$

and the generated law is $\tilde{p}_0^{(\gamma)} = \mathcal{N}(\mu_0 + \sigma_0 r_\gamma(\Lambda), \sigma_0^2 v_\gamma(\Lambda))$.

Proof. The drift is affine and the diffusion coefficient deterministic, so Gaussianity is preserved. With $m_\tau = \mathbb{E}Y_\tau$ and $q_\tau = \text{Var} Y_\tau$,

$$\frac{dm_\tau}{d\tau} = -\frac{1+\gamma}{2} \frac{g^2}{\alpha_\theta V} (m_\tau - \mu_\theta), \quad \frac{dq_\tau}{d\tau} = -(1+\gamma) \frac{g^2}{\alpha_\theta V} q_\tau + \gamma g^2.$$

Divide by $d\ell/d\tau = g^2/V$, use $1/\alpha_\theta = 1-a$ and $(\mu_\theta - \mu_0)/(\alpha_\theta \sigma_0) = u$ for the mean, and $dV/d\tau = -g^2$ for $v = q/V$. Variation of constants gives (2.10)–(2.11). $\square$

The two KL orientations between the generated and target laws are

$$(2.12) \quad h_q(\gamma) := \text{KL}(\tilde{p}_0^{(\gamma)} \| p_0) = \frac{1}{2}(v - \log v - 1 + r^2), \quad h_c(\gamma) := \text{KL}(p_0 \| \tilde{p}_0^{(\gamma)}) = \frac{1}{2}\left(\frac{1}{v} + \log v - 1 + \frac{r^2}{v}\right),$$

with $(r, v) = (r_\gamma(\Lambda), v_\gamma(\Lambda))$. We write h for either orientation when a statement holds for both.

2.4. Exact linearization. Define the two *response functionals*

$$(2.13) \quad L_\gamma[a; b_T] := e^{-\gamma\Lambda} b_T + (1+\gamma) \int_0^\Lambda e^{-\gamma(\Lambda-s)} a(s) \, ds,$$

$$(2.14) \quad M_\gamma[u; r_T] := e^{-\frac{1+\gamma}{2}\Lambda} r_T + \frac{1+\gamma}{2} \int_0^\Lambda e^{-\frac{1+\gamma}{2}(\Lambda-s)} u(s) \, ds,$$

and write $L_\gamma[a]$, $M_\gamma[u]$ when the prior is exact ($b_T = r_T = 0$). Let $\|f\| := \int_0^\Lambda |f(\ell)| \, d\ell$.

Lemma 2.2 (Exact linearization). *For every $\gamma \geq 0$ and all integrable a, u ,*

$$(2.15) \quad |v_\gamma(\Lambda) - 1 - L_\gamma[a; b_T]| \leq (1+\gamma)\|a\| e^{(1+\gamma)\|a\|} (|b_T| + (1+\gamma)\|a\|),$$

$$(2.16) \quad |r_\gamma(\Lambda) - M_\gamma[u; r_T]| \leq \frac{1+\gamma}{2}\|a\| e^{\frac{1+\gamma}{2}\|a\|} (|r_T| + \frac{1+\gamma}{2}\|u\|).$$

Proof. Put $y = v_\gamma - 1$. By (2.9), $y' = [-\gamma + (1+\gamma)a]y + (1+\gamma)a$ with $y(0) = b_T$, so

$$y(\Lambda) = e^{-\gamma\Lambda} e^{(1+\gamma)I_a(0)} b_T + (1+\gamma) \int_0^\Lambda e^{-\gamma(\Lambda-s)} e^{(1+\gamma)I_a(s)} a(s) \, ds.$$

Subtracting (2.13) and using $|e^x - 1| \leq |x|e^{|x|}$, $|I_a(s)| \leq \|a\|$ and $e^{-\gamma(\Lambda-s)} \leq 1$ gives (2.15). The same argument applied to (2.10) gives (2.16). $\square$

Proposition 2.3 (Second-order KL). *Let $\delta := \|a\| + \|u\| + |b_T| + |r_T|$. For every compact $\Gamma \subset [0, \infty)$ there is C_Γ such that, for $\delta \leq 1$ and $\gamma \in \Gamma$,*

$$(2.17) \quad \left| h(\gamma) - \frac{1}{4}L_\gamma[a; b_T]^2 - \frac{1}{2}M_\gamma[u; r_T]^2 \right| \leq C_\Gamma \delta^3$$

for both orientations $h \in \{h_q, h_c\}$. In particular there is no second-order cross term between the mean and shape modes.

Proof. By Lemma 2.2, $v - 1 = L + O(\delta^2)$ and $r = M + O(\delta^2)$, with $L, M = O(\delta)$. Both KL functions satisfy $h = \frac{1}{4}(v - 1)^2 + \frac{1}{2}r^2 + O(|v - 1|^3 + |v - 1|r^2)$ near $(v, r) = (1, 0)$. $\square$

Remark 2.4 (Why $L^1(d\ell)$ and not pointwise smallness). Near the data end, $d\ell = dy/(1 + y) \approx d\sigma^2/\sigma_0^2$. A score error with $|a| = O(1)$ on $\{\sigma \leq \sigma_\varepsilon\}$ contributes only $O(\sigma_\varepsilon^2/\sigma_0^2)$ to $\|a\|$. For the Gaussian-mixture checkpoints of Section 9, the root-mean-square envelope of $\log \alpha_\theta$ reaches about 2.5 at $\sigma = 0.002$, yet its $L^1(d\ell)$ norm is about 0.17, of which 43% comes from $\sigma < \sigma_0$. Earlier versions parametrized the shape error as $\alpha_\theta = e^{\varepsilon^A}$ and expanded pointwise in ε ; the coordinates (2.6) make that expansion unnecessary.

2.5. Two Gaussian tools. We repeatedly use two classical facts. If (X, Y) is a centered, nondegenerate bivariate Gaussian (or, more generally, centered elliptical without atom at the origin) with correlation ρ , then *Sheppard's orthant formula* holds [8]:

$$(2.18) \quad \mathbb{P}(XY > 0) = \frac{1}{2} + \frac{1}{\pi} \arcsin \rho.$$

If (U, V) is standard bivariate normal with correlation $\rho \in (-1, 1)$, its *product tail* is

$$(2.19) \quad T_\rho(z) := \mathbb{P}(UV > z) = 2 \int_0^\infty \phi(s) \Phi\left(\frac{\rho s - z/s}{\sqrt{1 - \rho^2}}\right) ds, \quad T_\rho(0) = \frac{1}{2} + \frac{1}{\pi} \arcsin \rho,$$

where ϕ and Φ are the standard normal density and distribution function; $z \mapsto T_\rho(z)$ is strictly decreasing. For noncentered pairs we use the standard bivariate normal distribution function $\Phi_2(h, k; \rho)$ [4].

3. THE STATIC BENCHMARK

This section records what happens when all misspecification parameters are constant in time, the setting of [7]. It is the zero-exponent member of the power-law families of Section 5, and it fixes the baseline against which time localization should be judged. We mainly ask the *local* question of whether a small amount of stochasticity helps, that is, whether $h'(0) < 0$, where the prime denotes the right derivative in γ . If $\mathbb{P}(h'(0) = 0) = 0$, dominated convergence gives $\lim_{\gamma \downarrow 0} \mathbb{P}(h(\gamma) < h(0)) = \mathbb{P}(h'(0) < 0)$. Random mean errors and Gamma priors are treated in Section A.

3.1. Exact formulas. Let $\alpha_\theta \equiv \alpha$, μ_θ , $\beta_T \equiv \beta$ and μ_T be constants. Proposition 2.1 gives

$$(3.1) \quad v(\gamma) = \frac{\gamma\alpha}{1 + \gamma - \alpha} + \left(\beta - \frac{\gamma\alpha}{1 + \gamma - \alpha}\right) \zeta_T^{(1+\gamma)/\alpha-1}, \quad r(\gamma) = \frac{\mu_\theta - \mu_0 + (\mu_T - \mu_\theta) \zeta_T^{(1+\gamma)/(2\alpha)}}{\sigma_0},$$

with the continuous extension $v = \beta + \gamma\Lambda$ at $1 + \gamma = \alpha$. Put $t = t(\alpha) := \Lambda(1/\alpha - 1)$ and $R_\alpha := \zeta_T^{1/(2\alpha)}$. At $\gamma = 0$,

$$(3.2) \quad v(0) = \beta e^{-t}, \quad v'(0) = \frac{\Lambda e^{-t}}{\alpha} (B_1(\alpha) - \beta), \quad r(0)r'(0) = -\frac{\Lambda R_\alpha}{2\alpha\sigma_0^2} ((1 - R_\alpha)X + R_\alpha Y)(Y - X),$$

where $X := \mu_\theta - \mu_0$, $Y := \mu_T - \mu_0$ and

$$(3.3) \quad B_0(\alpha) := e^t, \quad B_1(\alpha) := \alpha \frac{e^t - 1}{t}, \quad B_0(1) = B_1(1) = 1.$$

With the *shape contribution* $S(\alpha, \beta) := v'(0)(v(0) - 1)/(2v(0))$, differentiation of (2.12) gives

$$(3.4) \quad h'_q(0) = S + r(0)r'(0), \quad h'_c(0) = \frac{1}{v(0)} \left[S + r(0)r'(0) - \frac{v'(0)r(0)^2}{2v(0)} \right].$$

In the shape-only case $X = Y = 0$ the two orientations have the same sign.

Proposition 3.1 (Exact sign factorization). *For all $\alpha, \beta > 0$,*

$$(3.5) \quad S(\alpha, \beta) = -\frac{\Lambda e^{-t}}{2\alpha\beta} (\beta - B_0(\alpha))(\beta - B_1(\alpha)).$$

Hence $S < 0$ if and only if β lies outside the band $[B_-(\alpha), B_+(\alpha)]$, where B_- and B_+ are the smaller and the larger of B_0, B_1 . Moreover $B_1 < B_0$ for $\alpha < 1$ and $B_0 < B_1$ for $\alpha > 1$.

Proof. By (3.2), $v(0) - 1 = e^{-t}(\beta - B_0)$ and $v'(0) = \Lambda e^{-t}\alpha^{-1}(B_1 - \beta)$. The ordering follows from $B_1/B_0 = \alpha(1 - e^{-t})/t$, treating $t > 0$ and $t < 0$ separately. $\square$

Lemma 3.2 (Band probability). *If α and β are independent positive random variables and F_β is continuous, then*

$$(3.6) \quad \mathbb{P}(h'_q(0) < 0) = \mathbb{P}(h'_c(0) < 0) = 1 - \mathbb{E}[|F_\beta(B_1(\alpha)) - F_\beta(B_0(\alpha))|] \quad (X = Y = 0).$$

In particular the odds favor stochasticity locally if and only if $\mathbb{E}|F_\beta(B_1(\alpha)) - F_\beta(B_0(\alpha))| < 1/2$.

Proof. Condition on $\alpha = a$ in Proposition 3.1 and integrate. $\square$

3.2. Universal small-error limit. Put $x := \alpha - 1$ and $z := \beta - 1$. Expanding (3.2) gives $v(0) - 1 = z + \Lambda x + O(|x|^2 + |z|^2)$ and

$$(3.7) \quad S(\alpha, \beta) = \frac{\Lambda}{4} (\Lambda x + z)((2 - \Lambda)x - 2z) + O((|x| + |z|)^3).$$

Proposition 3.3 (Distribution-free local limit). *Let $(\alpha_\varepsilon, \beta_\varepsilon)$ be positive random vectors with $((\alpha_\varepsilon - 1)/\varepsilon, (\beta_\varepsilon - 1)/\varepsilon) \Rightarrow (X_1, X_2)$, and put $U_\Lambda := \Lambda X_1 + X_2$ and $W_\Lambda := (2 - \Lambda)X_1 - 2X_2$. If $\mathbb{P}(U_\Lambda W_\Lambda = 0) = 0$, then for both orientations, with exact means,*

$$\lim_{\varepsilon \downarrow 0} \mathbb{P}(h'(0) < 0) = \mathbb{P}(U_\Lambda W_\Lambda < 0).$$

If (X_1, X_2) is centered elliptical with scatter matrix $\begin{pmatrix} s_{11} & s_{12} \\ s_{12} & s_{22} \end{pmatrix} > 0$, the limit equals $\frac{1}{2} + \frac{1}{\pi} \arcsin \kappa_\Lambda$ with

$$(3.8) \quad \kappa_\Lambda = \frac{\Lambda(\Lambda - 2)s_{11} + (3\Lambda - 2)s_{12} + 2s_{22}}{\sqrt{(\Lambda^2 s_{11} + 2\Lambda s_{12} + s_{22})((2 - \Lambda)^2 s_{11} - 4(2 - \Lambda)s_{12} + 4s_{22})}}.$$

In the diagonal case with $\eta := s_{22}/s_{11}$, the limit exceeds $1/2$ if and only if $\Lambda(\Lambda - 2) + 2\eta > 0$. This holds for every $\Lambda > 0$ when $\eta > 1/2$. For $\eta = 1$ the limiting odds are at least 0.5999, with the minimum at $\Lambda = \sqrt{10} - 2$.

Proof. The remainder in (3.7) is $o_P(\varepsilon^2)$; apply the continuous mapping theorem, $v(0) \rightarrow 1$ in probability, and (2.18) to (U_Λ, W_Λ) , noting that $\mathbb{P}(UW < 0) = \frac{1}{2} - \frac{1}{\pi} \arcsin \text{Corr}(U, W)$. $\square$

Proposition 3.4 (Exact e^{-2} threshold with exact prior shape). *Let $\beta = 1$ and $R(t) := (e^t - 1 - t)/t^2$, which is strictly increasing with $R(0) = 1/2$. Let t^* solve $R(t^*) = 1/\Lambda$ and put $\alpha^*(\Lambda) := \Lambda/(\Lambda + t^*)$. Then $\{\alpha > 0 : S(\alpha, 1) < 0\} = (0, \alpha^*(\Lambda)) \setminus \{1\}$, and $\alpha^* < 1$, $= 1$ or > 1 according as $\Lambda < 2$, $= 2$ or > 2 . Consequently, for every prior on α with continuous, strictly increasing distribution function and median one,*

$$\mathbb{P}(S(\alpha, 1) < 0) \gtrless \frac{1}{2} \iff \zeta_T \lesseqgtr e^{-2}.$$

Proof. One checks $B_1(\alpha) - 1 = t[\Lambda R(t) - 1]/(\Lambda + t)$, so for $\alpha \neq 1$, $S(\alpha, 1) < 0$ iff $R(t) > 1/\Lambda$. Since t decreases in α and R increases, the claim follows. $\square$

Proposition 3.5 (Strong-noising endpoint). *For any fixed joint law of (α, β) on $(0, \infty)^2$, with exact means,*

$$\lim_{\zeta_T \downarrow 0} \mathbb{P}(h'(0) < 0) = 1 - \mathbb{P}(\alpha = 1, \beta = 1).$$

The same limit holds with independent Gaussian mean errors whenever $\mathbb{P}(\alpha = 1) = 0$ (Proposition A.3).

Proof. For $\alpha < 1$ both boundaries B_0, B_1 tend to $+\infty$ as $\Lambda \rightarrow \infty$, and for $\alpha > 1$ both tend to 0. At $\alpha = 1$, $S(1, \beta) = -\Lambda(\beta - 1)^2/(2\beta)$. Conclude by dominated convergence. $\square$

At the weak-noising endpoint $\zeta_T \uparrow 1$ the band tends to $[\min(1, \alpha), \max(1, \alpha)]$. For independent and identically distributed α, β with continuous law and $q := F(1)$, the limiting odds are $\frac{1}{2} + q(1 - q)$, which equals $3/4$ at median one.

Remark 3.6 (What the static benchmark predicts at practical horizons). At $\Lambda \approx 10$ – 11 , one has $\zeta_T \approx 2 \times 10^{-5}$ to 4×10^{-5} . Proposition 3.5 then says that, for any atomless static prior, a small amount of stochasticity helps with probability close to one. Static mean errors are nearly invisible at this horizon, since they enter $r'(0)$ only through $R_\alpha \approx e^{-\Lambda/2} \approx 4 \times 10^{-3}$. Both statements are artifacts of spreading the error uniformly over the clock. In the language of Section 4, a constant shape profile has $L_0[a] = \Lambda a$, which grows with Λ , whereas $L_\gamma[a] \leq (1 + \gamma)a/\gamma$ stays bounded. The measured networks of Section 9 concentrate their error near the data end, where this argument fails.

4. STOCHASTICITY AS A TEMPORAL FILTER

4.1. Leading-order comparison. Let $a = \varepsilon A$ and $u = \varepsilon U$ with fixed integrable profiles and an exact prior. By Proposition 2.3,

$$(4.1) \quad h(\gamma) - h(0) = \varepsilon^2 Q_\gamma[A, U] + O(\varepsilon^3), \quad Q_\gamma[A, U] := \frac{1}{4}(L_\gamma[A]^2 - L_0[A]^2) + \frac{1}{2}(M_\gamma[U]^2 - M_0[U]^2),$$

for both KL orientations, uniformly for γ in compact sets. Thus if $Q_\gamma \neq 0$, the sign of $h(\gamma) - h(0)$ is that of Q_γ for all sufficiently small $\varepsilon \neq 0$. In the shape-only case, stochasticity helps if and only if $|L_\gamma[A]| < |L_0[A]|$. Equation (4.1) makes explicit that “SDE-favorable” and “ODE-favorable” are properties of the temporal composition of the error: an early shape error for which stochasticity is corrective can coexist with a late mean error for which it is harmful.

4.2. Kernels in physical units. In the relative-noise variable $y = \sigma^2/\sigma_0^2$ one has $dd = dy/(1 + y)$ and $e^{-\gamma d} = (1 + y)^{-\gamma}$, so that, with $y_T := e^\Lambda - 1$,

$$(4.2) \quad L_\gamma[a] = (1 + \gamma) \int_0^{y_T} a(1 + y)^{-1-\gamma} dy, \quad M_\gamma[u] = \frac{1 + \gamma}{2} \int_0^{y_T} u(1 + y)^{-\frac{3+\gamma}{2}} dy.$$

The ODE–SDE differences are integrals against signed kernels,

$$(4.3) \quad \begin{aligned} L_\gamma - L_0 &= \int_0^\Lambda k_\gamma(d) a \, dd, & k_\gamma(d) &:= (1 + \gamma)e^{-\gamma d} - 1, \\ M_\gamma - M_0 &= \frac{1}{2} \int_0^\Lambda e^{-d/2} k_\gamma^u(d) u \, dd, & k_\gamma^u(d) &:= (1 + \gamma)e^{-\gamma d/2} - 1. \end{aligned}$$

Proposition 4.1 (Crossover noise levels). *Fix $\gamma > 0$. The kernel k_γ is positive for $d < d_\gamma := \log(1 + \gamma)/\gamma$ and negative for $d > d_\gamma$, and k_γ^u changes sign at $2d_\gamma$. In terms of the added noise level these crossovers are*

$$(4.4) \quad \sigma_a^*(\gamma) = \sigma_0 \sqrt{(1 + \gamma)^{1/\gamma} - 1}, \quad \sigma_u^*(\gamma) = \sigma_0 \sqrt{(1 + \gamma)^{2/\gamma} - 1}.$$

Both are strictly decreasing in γ . As $\gamma \downarrow 0$ they tend to $\sigma_0 \sqrt{e - 1} \approx 1.31 \sigma_0$ and $\sigma_0 \sqrt{e^2 - 1} \approx 2.53 \sigma_0$. At $\gamma = 1$ they equal σ_0 and $\sqrt{3} \sigma_0$, and they tend to 0 as $\gamma \rightarrow \infty$. Consequently, for a nonnegative shape profile $A \not\equiv 0$ (exact mean):

- (i) stochasticity helps at leading order if and only if $\int_0^\Lambda k_\gamma A \, dd < 0$;
- (ii) if $\text{supp } A \subset \{\sigma \geq \sigma_a^*(\gamma)\}$, stochasticity helps; if $\text{supp } A \subset \{\sigma \leq \sigma_a^*(\gamma)\}$, it hurts.

The analogous statements hold for a nonnegative mean profile with σ_u^* .

Proof. $k_\gamma(d) > 0$ iff $e^{\gamma d} < 1 + \gamma$. Since $V/\sigma_0^2 = e^d$, the crossover is $1 + \sigma^2/\sigma_0^2 = (1 + \gamma)^{1/\gamma}$. The function $\gamma \mapsto \log(1 + \gamma)/\gamma$ is strictly decreasing from 1 to 0. For $A \geq 0$ both $L_\gamma[A]$ and $L_0[A]$ are positive, so $|L_\gamma| < |L_0|$ iff $L_\gamma < L_0$. $\square$

In words, stochastic sampling *amplifies* shape error below $\sigma_a^*(\gamma)$ and *damps* it above; for the standard reverse SDE the dividing line is the data scale. The mean mode is already contracted by the ODE, which is why its crossover sits higher.

4.3. Infinitesimal stochasticity: centroid criteria. Differentiating (2.13)–(2.14) at $\gamma = 0$,

$$(4.5) \quad \partial_\gamma L_\gamma[A]|_{\gamma=0} = \int_0^\Lambda (1 - d) A \, dd, \quad \partial_\gamma M_\gamma[U]|_{\gamma=0} = \frac{1}{4} \int_0^\Lambda e^{-d/2} (2 - d) U \, dd,$$

so that $h'(0) = \varepsilon^2 (\frac{1}{2} L_0[A] \partial_\gamma L_0[A] + M_0[U] \partial_\gamma M_0[U]) + O(\varepsilon^3)$.

Theorem 4.2 (Centroid criteria). *For nonnegative profiles $A, U \not\equiv 0$ define the centroids*

$$\bar{d}_A := \frac{\int_0^\Lambda d A \, dd}{\int_0^\Lambda A \, dd}, \quad \bar{d}_U^u := \frac{\int_0^\Lambda d e^{-d/2} U \, dd}{\int_0^\Lambda e^{-d/2} U \, dd}.$$

With exact mean, $h'(0) < 0$ at leading order if and only if $\bar{d}_A > 1$. With exact shape, $h'(0) < 0$ at leading order if and only if $\bar{d}_U^u > 2$.

Proof. Factor $\int A$ (respectively $\int e^{-d/2} U$) out of (4.5). $\square$

These thresholds are the $\gamma \downarrow 0$ limits of the crossovers, $d_\gamma \rightarrow 1$ and $2d_\gamma \rightarrow 2$. The constant profile $A \equiv 1$ has $\bar{d}_A = \Lambda/2$ and recovers the static threshold $\Lambda > 2$, that is $\zeta_T < e^{-2}$ (Proposition 3.4).

4.4. Prior mismatch and stochasticity schedules. By (2.13)–(2.14), prior errors are boundary modes concentrated at the start of sampling. The prior-variance error b_T is weighted by $e^{-\gamma\Lambda}$; in particular it is not damped at all by the ODE, which is the mechanism behind [Proposition 3.5](#). The prior-mean error is weighted by $e^{-(1+\gamma)\Lambda/2}$. For EDM-like priors, $b_T \approx -\sigma_0^2/\sigma(T)^2$ is negligible.

For a time-dependent schedule $\gamma = \gamma(\ell)$, the same derivation gives

$$(4.6) \quad L_{\gamma(\cdot)}[a] = \int_0^\Lambda (1 + \gamma(s)) e^{-\int_s^\Lambda \gamma(r) dr} a(s) ds, \quad M_{\gamma(\cdot)}[u] = \int_0^\Lambda \frac{1 + \gamma(s)}{2} e^{-\int_s^\Lambda \frac{1+\gamma(r)}{2} dr} u(s) ds.$$

Choosing the schedule is thus a filter-design problem for the error profile; see [Section 10](#).

5. POWER-LAW PROFILES

Throughout this section profiles are power laws in the added noise level, normalized at the data scale,

$$(5.1) \quad A_p(\sigma) := \left(\frac{\sigma_0}{\sigma}\right)^p = y^{-p/2}, \quad U_{p_u}(\sigma) := \left(\frac{\sigma_0}{\sigma}\right)^{p_u} = y^{-p_u/2},$$

and we write $q := p/2$ and $q_u := p_u/2$. Let $B(x; \alpha, \beta) := \int_0^x w^{\alpha-1} (1-w)^{\beta-1} dw$ be the incomplete Beta function and $B(\alpha, \beta) = B(1; \alpha, \beta)$; ψ is the digamma function.

5.1. Incomplete-Beta representation.

Lemma 5.1. *Let $w_T := 1 - \zeta_T = 1 - e^{-\Lambda}$. For $p < 2$ and $p_u < 2$,*

$$(5.2) \quad L_\gamma[A_p] = (1 + \gamma) B(w_T; 1 - q, q + \gamma), \quad M_\gamma[U_{p_u}] = \frac{1 + \gamma}{2} B(w_T; 1 - q_u, q_u + \frac{1+\gamma}{2}).$$

As $\Lambda \rightarrow \infty$ these converge to the complete Beta functions $(1 + \gamma)B(1 - q, q + \gamma)$ and $\frac{1+\gamma}{2}B(1 - q_u, q_u + \frac{1+\gamma}{2})$. This requires $q + \gamma > 0$ (so $p > 0$ when $\gamma = 0$) and $q_u + \frac{1+\gamma}{2} > 0$ (so $p_u > -1$ when $\gamma = 0$).

Proof. Substitute $w = y/(1 + y) = 1 - e^{-d} = \sigma^2/V$ in (4.2): $y^{-q}(1 + y)^{-1-\gamma} dy = w^{-q}(1 - w)^{q+\gamma-1} dw$ and $y^{-q}(1 + y)^{-(3+\gamma)/2} dy = w^{-q}(1 - w)^{q+(\gamma-1)/2} dw$. $\square$

The variable $w = \sigma^2/V$ is the fraction of the current variance that is noise. Because the response functionals are linear, sums of powers give sums of Beta functions. A continuous piecewise power law, which is the form of the empirical fits in [Section 9](#), also gives a closed form: if $A = c_j(\sigma_0/\sigma)^{p_j}$ on (σ_{j-1}, σ_j) , then

$$(5.3) \quad L_\gamma[A] = (1 + \gamma) \sum_j c_j \left[B(w_j; 1 - q_j, q_j + \gamma) - B(w_{j-1}; 1 - q_j, q_j + \gamma) \right], \quad w_j := \frac{\sigma_j^2}{\sigma_0^2 + \sigma_j^2},$$

and similarly for M_γ . More generally, for a superposition $A = \int (\sigma_0/\sigma)^p \rho(dp)$ of powers, $L_\gamma[A] = \int (1 + \gamma)B(1 - p/2, p/2 + \gamma) \rho(dp)$: the response functionals are Mellin transforms of the kernels.

Remark 5.2 (Stopping at $\sigma_{\min} > 0$). Samplers stop at a positive noise level $\sigma_{\min}$ and are compared with $p_{\sigma_{\min}}$. This is the same model with σ_0^2 replaced by $V_{\min} = \sigma_0^2 + \sigma_{\min}^2$ and y by $(\sigma^2 - \sigma_{\min}^2)/V_{\min}$. For $p < 2$ the change affects the functionals by $O((\sigma_{\min}/\sigma_0)^{2-p})$, which is negligible in the applications ($\sigma_{\min}/\sigma_0 \approx 6 \times 10^{-3}$).

γ	0^+	0.1	0.25	0.5	1	2	3	5	10
$p_c(\gamma)$ (Theorem 5.3)	1.180	1.153	1.117	1.069	1.000	0.915	0.862	0.797	0.717
$p_{\text{wn}}(\gamma)$ (Theorem 7.2)	1.000	0.972	0.936	0.889	0.823	0.744	0.696	0.638	0.568

TABLE 1. Critical shape exponents. p_c : deterministic power-law profile. p_{wn} : exponent at which the white-noise Gaussian-process odds cross 1/2.

5.2. Shape error: a γ -dependent critical exponent.

Theorem 5.3 (Critical shape exponent). *For $\gamma > 0$ and $0 < p < 2$ define*

$$(5.4) \quad \mathcal{R}_\gamma(p) := \lim_{\Lambda \rightarrow \infty} \frac{L_\gamma[A_p]}{L_0[A_p]} = \frac{(1 + \gamma) \Gamma(\frac{p}{2} + \gamma)}{\Gamma(\frac{p}{2}) \Gamma(1 + \gamma)}.$$

Then $\mathcal{R}_\gamma$ is strictly increasing on $(0, 2)$, with $\mathcal{R}_\gamma(0^+) = 0$ and $\mathcal{R}_\gamma(2^-) = 1 + \gamma$. Hence there is a unique $p_c(\gamma) \in (0, 2)$ with $\mathcal{R}_\gamma(p_c) = 1$, and for shape-only power-law error of small amplitude and sufficiently large horizon,

$$p < p_c(\gamma) \implies h(\gamma) < h(0), \quad p > p_c(\gamma) \implies h(\gamma) > h(0).$$

(For each compact $K \subset (0, 2) \setminus \{p_c(\gamma)\}$ there is Λ_0 such that the statement holds for all $p \in K$ and $\Lambda \geq \Lambda_0$.) Moreover:

- (i) $\mathcal{R}_1(p) = p$, so $p_c(1) = 1$ exactly; for integer $\gamma = n$, $\mathcal{R}_n(p) = (n + 1) (p/2)_n / n!$, with $(x)_n$ the rising factorial;
- (ii) $\mathcal{R}_\gamma(p) = 1 + \gamma(1 + \psi(\frac{p}{2}) - \psi(1)) + O(\gamma^2)$, so $p_c(\gamma) \rightarrow p_c(0^+)$ as $\gamma \downarrow 0$, where $\psi(p_c(0^+)/2) = -1 - \gamma_E$ (γ_E is Euler's constant) and $p_c(0^+) \approx 1.1801$. The condition $p < p_c(0^+)$ is exactly the local criterion $h'(0) < 0$ of Theorem 4.2 for this family;
- (iii) $p_c(\gamma) \rightarrow 0$ as $\gamma \rightarrow \infty$.

Proof. By Lemma 5.1, $L_\gamma/L_0 \rightarrow (1 + \gamma)B(1 - q, q + \gamma)/B(1 - q, q)$, which simplifies to (5.4). Next, $\partial_q \log \mathcal{R}_\gamma = \psi(q + \gamma) - \psi(q) > 0$ because ψ is strictly increasing. As $q \downarrow 0$, $\Gamma(q) \rightarrow \infty$; at $q = 1$ the ratio is $1 + \gamma$. The sign statement for large Λ follows from uniform convergence on compacts, and Eq. (4.1) with $U = 0$ gives the comparison. Item (i) is $\Gamma(q + n)/\Gamma(q) = (q)_n$. Item (ii) follows from $\Gamma(q + \gamma)/\Gamma(q) = 1 + \gamma\psi(q) + O(\gamma^2)$ and the implicit function theorem, since $\psi' > 0$. For (iii), $\mathcal{R}_\gamma(p) \sim \gamma^{p/2}/\Gamma(p/2) \rightarrow \infty$ for each fixed $p > 0$. $\square$

Conjecture 5.4. $\gamma \mapsto p_c(\gamma)$ is strictly decreasing on $(0, \infty)$. (This holds numerically; see Table 1 and Fig. 1.)

Proposition 5.5 (Optimal and maximal useful stochasticity). *Let $0 < p < 2$ and $\Lambda \rightarrow \infty$. The map $\gamma \mapsto L_\gamma[A_p]$ is continuous on $[0, \infty)$ and tends to $+\infty$ as $\gamma \rightarrow \infty$, since $L_\gamma \sim \Gamma(1 - \frac{p}{2})\gamma^{p/2}$. Hence:*

- (i) *a minimizer $\gamma^*(p)$ of L_γ over constant $\gamma \geq 0$ exists. It is positive if $p < p_c(0^+)$ (and numerically $\gamma^* = 0$ for $p \geq p_c(0^+)$), and every positive minimizer satisfies*
- $$(5.5) \quad \psi(1 + \gamma) - \psi(\frac{p}{2} + \gamma) = \frac{1}{1 + \gamma};$$

- (ii) *the set $\{\gamma > 0 : L_\gamma < L_0\}$ is bounded; its supremum $\gamma_c(p)$ is the largest useful stochasticity. If ?? 5.4 holds, the set is the interval $(0, \gamma_c(p))$, and $p_c(\gamma_c(p)) = p$.*

Proof. Equation (5.5) is $\partial_\gamma \log[(1 + \gamma)\Gamma(q + \gamma)/\Gamma(1 + \gamma)] = 0$. Positivity of γ^* for $p < p_c(0^+)$ follows from Theorem 5.3(ii), since then $\partial_\gamma L_\gamma|_{\gamma=0} < 0$. Boundedness of the set in (ii) follows from the growth of L_γ . $\square$

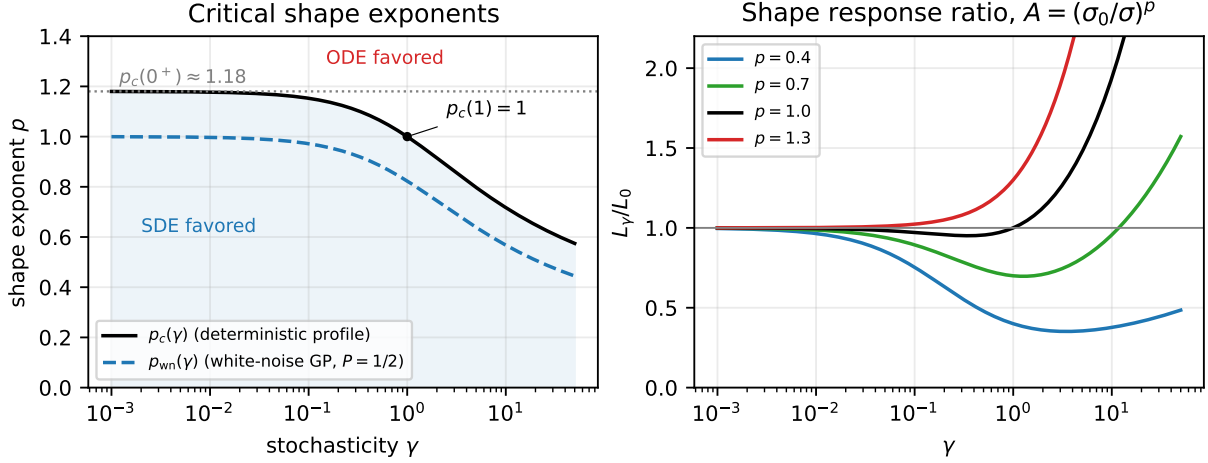


FIGURE 1. Left: critical shape exponent $p_c(\gamma)$ for deterministic power-law profiles (solid) and the exponent $p_{\text{wn}}(\gamma)$ at which the white-noise Gaussian-process odds equal 1/2 (dashed). Right: L_γ/L_0 as a function of γ for several exponents ($\Lambda \rightarrow \infty$).

p	0.2	0.4	0.6	0.8	1.0	1.1
$\gamma^*(p)$	8.48	3.46	1.76	0.894	0.351	0.145
$h(\gamma^*)/h(0) = (L_{\gamma^*}/L_0)^2$	0.021	0.124	0.341	0.639	0.904	0.980
$\gamma_c(p)$	$> 10^8$	2.0×10^3	35.5	4.90	1	0.333

TABLE 2. Optimal constant stochasticity γ^* , the corresponding leading-order KL ratio, and the largest useful stochasticity γ_c for shape-only error $(\sigma_0/\sigma)^p$, $\Lambda \rightarrow \infty$.

5.3. Why universality is lost: comparison with the exponential family. Earlier versions of this work used the family $A_\kappa(d) = e^{-\kappa d} = (\sigma_0^2/V)^\kappa = (1+y)^{-\kappa}$.

Proposition 5.6 (Exponential family). *For $\gamma, \kappa > 0$, as $\Lambda \rightarrow \infty$, $L_0[A_\kappa] \rightarrow 1/\kappa$ and $L_\gamma[A_\kappa] \rightarrow (1+\gamma)/(\gamma+\kappa)$. Hence $L_\gamma < L_0$ if and only if $\kappa < 1$, for every $\gamma > 0$. At finite Λ with $(1+\gamma)(1-e^{-\gamma\Lambda}) < \gamma\Lambda$, there is a unique crossing $\kappa_c(\gamma, \Lambda)$, which converges to 1 exponentially fast.*

Proof. The limits are elementary. The inequality $(1+\gamma)/(\gamma+\kappa) < 1/\kappa$ is equivalent to $\kappa < 1$. At finite Λ , $L_\gamma/L_0 = (1+\gamma)\mathbb{E}_{\nu_\kappa}[e^{-\gamma d}]$, where $\nu_\kappa \propto e^{-\kappa d} dd$ on $[0, \Lambda]$; this is strictly increasing in κ . $\square$

For $\sigma \gg \sigma_0$, $(\sigma_0/\sigma)^p \approx (1+y)^{-p/2}$, so the two families agree at high noise with $\kappa = p/2$; the exponential threshold $\kappa_c = 1$ corresponds to $p = 2$. They differ only for $\sigma \lesssim \sigma_0$. There the power law has an integrable singularity $d^{-p/2}$ at the data end, exactly where $e^{-\gamma d} \approx 1$ and the stochastic sampler weighs error by $1+\gamma$. This near-end mass drives $L_\gamma \sim \Gamma(1 - \frac{p}{2})\gamma^{p/2} \rightarrow \infty$, whereas $L_\gamma[A_\kappa] \rightarrow 1$. The universality of $\kappa_c = 1$ was therefore a consequence of assuming a profile bounded at the data end. For realistic profiles the threshold moves down to $p_c(\gamma) \in (0, 1.18)$ and depends on γ .

Remark 5.7 ($p \geq 2$). For $p \geq 2$ the functionals diverge at the data end and the stopping level of Remark 5.2 is essential. Both L_0 and L_γ are then dominated by $d \lesssim \sigma_{\min}^2/\sigma_0^2$, where $k_\gamma \approx \gamma$. Hence $L_\gamma/L_0 \rightarrow 1+\gamma$ as $\sigma_{\min} \rightarrow 0$, and the ODE is preferred.

Remark 5.8 (Exponents of the score-error norm). By (2.7), $\mathbb{E}_{p_t}[\epsilon_t^{\text{shape}}(X)^2] = a^2/V$ and $\epsilon_t^{\text{mean}} = \sigma_0 u/V$. For $a \propto \sigma^{-p}$ and $u \propto \sigma^{-p_u}$, the squared error therefore scales as σ^{-2p} and σ^{-2p_u} for $\sigma \ll \sigma_0$, and as σ^{-2p-2} and σ^{-2p_u-4} for $\sigma \gg \sigma_0$. In particular, the constant-mean threshold ($p_u = 0$) corresponds to $\mathbb{E}|\epsilon|^2 \propto \sigma^{-4}$ at high noise, not σ^{-2} as labelled in an earlier figure.

5.4. Mean error: a universal threshold.

Theorem 5.9 (Critical mean exponent). *For $\gamma > 0$ and $-1 < p_u < 2$,*

$$(5.6) \quad \mathcal{R}_\gamma^u(p_u) := \lim_{\Lambda \rightarrow \infty} \frac{M_\gamma[U_{p_u}]}{M_0[U_{p_u}]} = \frac{(1+\gamma)\Gamma(\frac{3}{2})\Gamma(\frac{p_u}{2} + \frac{1+\gamma}{2})}{\Gamma(\frac{3+\gamma}{2})\Gamma(\frac{p_u+1}{2})}$$

is strictly increasing with $\mathcal{R}_\gamma^u(0) = 1$. Hence the critical mean exponent is $p_u^c = 0$ for every $\gamma > 0$: mean error that grows toward the data end favors the ODE, and mean error that decays toward it favors stochasticity. At finite Λ the constant profile satisfies $M_\gamma - M_0 = e^{-\Lambda/2} - e^{-(1+\gamma)\Lambda/2} > 0$, an exponentially small preference for the ODE.

Proof. $\partial_{q_u} \log \mathcal{R}_\gamma^u = \psi(q_u + \frac{1+\gamma}{2}) - \psi(q_u + \frac{1}{2}) > 0$. At $q_u = 0$, $\Gamma(\frac{3+\gamma}{2}) = \frac{1+\gamma}{2}\Gamma(\frac{1+\gamma}{2})$ and $\Gamma(\frac{3}{2})/\Gamma(\frac{1}{2}) = \frac{1}{2}$ give $\mathcal{R}_\gamma^u(0) = 1$. $\square$

The constant profile belongs to both the power-law and the exponential families, which is why the mean threshold is universal in both. The same threshold $\kappa_u^c = 0$ holds for exponential mean profiles.

5.5. Combined phase diagram. Let $a = \varepsilon_a A_{p_a}$ and $u = \varepsilon_u U_{p_u}$, with amplitudes measured at $\sigma = \sigma_0$. By (4.1) and Lemma 5.1, the leading-order comparison is

$$(5.7) \quad h(\gamma) - h(0) \approx \frac{\varepsilon_a^2}{4} \Delta L_\gamma(p_a) + \frac{\varepsilon_u^2}{2} \Delta M_\gamma(p_u), \quad \Delta L_\gamma := L_\gamma^2 - L_0^2, \quad \Delta M_\gamma := M_\gamma^2 - M_0^2,$$

all explicit Beta functions. Four regimes follow. For $p_a < p_c(\gamma)$ and $p_u < 0$ stochasticity helps; for $p_a > p_c(\gamma)$ and $p_u > 0$ it hurts. In the mixed regime $p_a < p_c(\gamma)$, $p_u > 0$, stochasticity helps if and only if

$$(5.8) \quad \frac{\varepsilon_u^2}{\varepsilon_a^2} < c_\gamma(p_a, p_u)^2 := \frac{-\Delta L_\gamma(p_a)}{2 \Delta M_\gamma(p_u)},$$

and symmetrically in the other mixed regime. Figure 2 shows the resulting diagrams. The mixed regime is the one suggested by the measurements of Section 9.

6. THE ODDS I: RANDOM AMPLITUDES AND RANDOM EXPONENTS

We now regard the error profile of a trained network as random, and ask for the probability that stochasticity improves the generated law. All probabilities below are taken with respect to the law of the profile. The KL divergences are computed conditionally on the profile; they are not divergences of the unconditional Gaussian-mixture output.

6.1. A single scalar amplitude gives degenerate odds.

Lemma 6.1. *Let $a = sZA$ and $u = sZU$ with deterministic profiles A, U , a scalar random variable Z with $\mathbb{P}(Z = 0) = 0$, and $s \downarrow 0$. If $Q_\gamma[A, U] \neq 0$, then $\mathbb{P}(h(\gamma) < h(0)) \rightarrow \mathbf{1}\{Q_\gamma[A, U] < 0\}$.*

Proof. By (4.1), $h(\gamma) - h(0) = s^2 Z^2 Q_\gamma + O(s^3 |Z|^3)$, whose sign is that of Q_γ whenever $s|Z|$ is small; the event $\{s|Z| > \eta\}$ has vanishing probability. $\square$

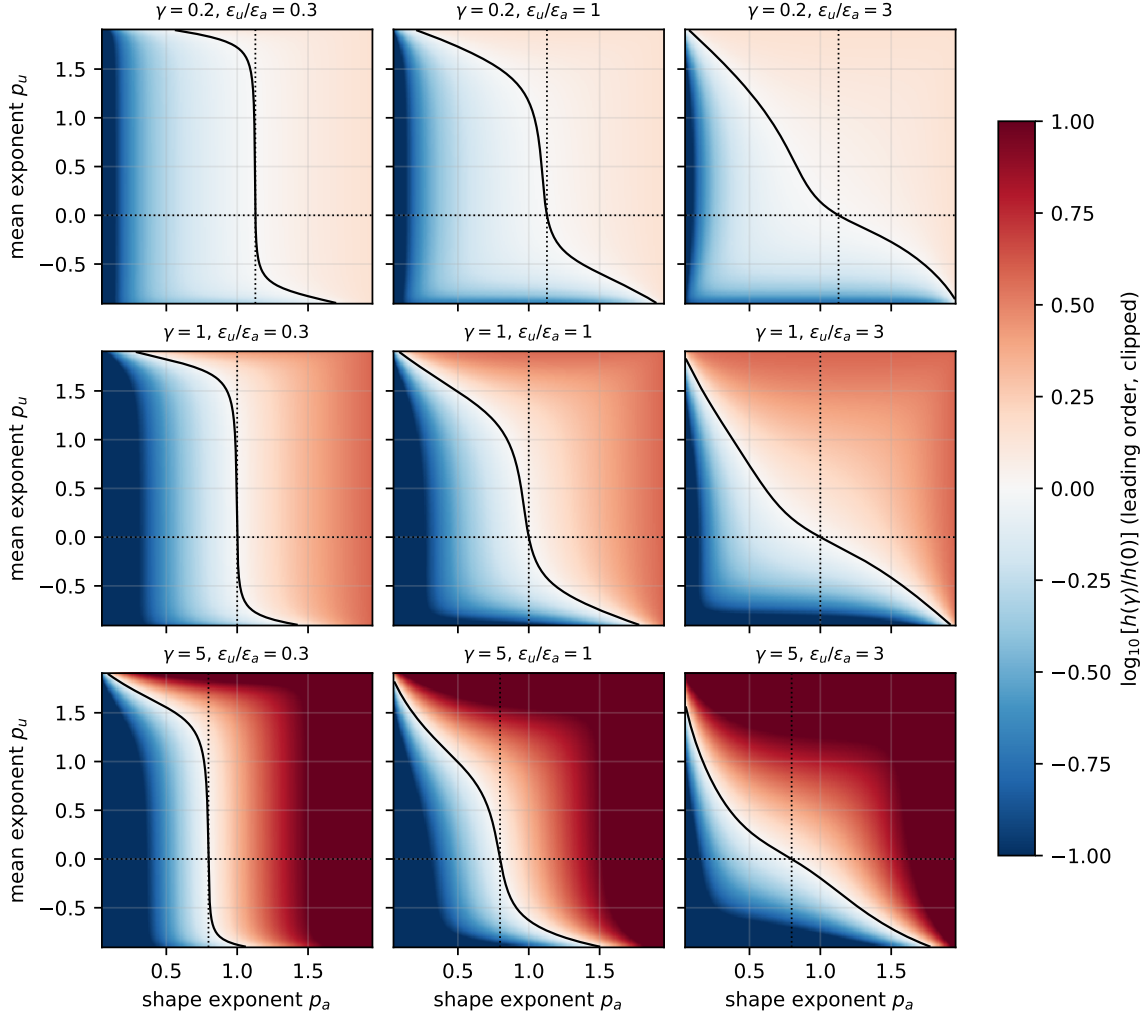


FIGURE 2. Leading-order $\log_{10}[h(\gamma)/h(0)]$ for power-law shape and mean profiles ($\Lambda \rightarrow \infty$), for $\gamma \in \{0.2, 1, 5\}$ (rows) and amplitude ratios $\varepsilon_u/\varepsilon_a \in \{0.3, 1, 3\}$ (columns). Blue: stochastic sampling favored; red: ODE favored. Solid: equal KL. Dotted: $p_a = p_c(\gamma)$ and $p_u = 0$.

Thus a single random amplitude multiplying a fixed temporal profile, whatever its law (Gaussian, lognormal or other), cannot produce nontrivial odds at leading order. Nontrivial odds require at least one of the following: several independent modes, a random temporal shape, or finite amplitude (Section 8). We treat the first two.

6.2. Independent mode amplitudes.

Theorem 6.2 (Two-amplitude odds). *Let $a = sZ_a A$ and $u = sZ_u U$ with deterministic profiles and a random vector (Z_a, Z_u) satisfying $\mathbb{P}(Z_a^2 \Delta L_\gamma / 4 + Z_u^2 \Delta M_\gamma / 2 = 0) = 0$, where $\Delta L_\gamma := L_\gamma[A]^2 - L_0[A]^2$ and $\Delta M_\gamma := M_\gamma[U]^2 - M_0[U]^2$. Then, as $s \downarrow 0$,*

$$\mathbb{P}(h(\gamma) < h(0)) \longrightarrow \mathbb{P}\left(\frac{1}{4}Z_a^2 \Delta L_\gamma + \frac{1}{2}Z_u^2 \Delta M_\gamma < 0\right).$$

The limit is 1 if $\Delta L_\gamma, \Delta M_\gamma \leq 0$ (not both zero) and 0 if both are nonnegative. In the mixed case $\Delta L_\gamma < 0 < \Delta M_\gamma$, with $c := \sqrt{-\Delta L_\gamma / (2\Delta M_\gamma)}$, it equals $\mathbb{P}(|Z_u| < c|Z_a|)$, and:

(i) if (Z_a, Z_u) is centered Gaussian (or centered elliptical) with scales s_a, s_u and correlation r ,

$$(6.1) \quad \mathbb{P}(h(\gamma) < h(0)) \rightarrow \frac{1}{2} - \frac{1}{\pi} \arcsin \frac{s_u^2 - c^2 s_a^2}{\sqrt{(s_u^2 + c^2 s_a^2)^2 - 4c^2 r^2 s_a^2 s_u^2}};$$

for $r = 0$ this is $\frac{2}{\pi} \arctan(c s_a/s_u)$;

(ii) if $(\log|Z_a|, \log|Z_u|)$ is Gaussian with means (m_a, m_u) and $\tau^2 := \text{Var}(\log|Z_u| - \log|Z_a|) > 0$,

$$(6.2) \quad \mathbb{P}(h(\gamma) < h(0)) \rightarrow \Phi\left(\frac{\log c - (m_u - m_a)}{\tau}\right).$$

The other mixed case $\Delta M_\gamma < 0 < \Delta L_\gamma$ is obtained by exchanging the roles of the two modes.

Proof. The first display follows as in [Lemma 6.1](#). In the mixed case the event is $Z_u^2 < c^2 Z_a^2$, that is, $(Z_u - cZ_a)(Z_u + cZ_a) < 0$. For (i), apply (2.18) to this centered pair. Its covariance is $s_u^2 - c^2 s_a^2$ and its variances are $s_u^2 + c^2 s_a^2 \mp 2crs_a s_u$. For $r = 0$, write $\kappa = cs_a/s_u = \tan \theta$; then $\arcsin \frac{1-\kappa^2}{1+\kappa^2} = \frac{\pi}{2} - 2\theta$. For (ii), the event is $\log|Z_u| - \log|Z_a| < \log c$. $\square$

At leading order only the laws of $|Z_a|$ and $|Z_u|$ matter. In particular, a Gaussian Z_a in $a = sZ_a A$ already makes $\alpha_\theta = 1/(1-a)$ approximately lognormal at each time. Lognormal magnitudes, as in (ii), are natural when checkpoint amplitudes spread over orders of magnitude; they turn the odds into a probit law in $\log c$. Since $c = c_\gamma(p_a, p_u)$ is explicit ([Eq. \(5.8\)](#)), [Theorem 6.2](#) gives the odds as a function of (p_a, p_u, γ) and of the amplitude law.

Remark 6.3 (Relation with the static layer). [Proposition 3.3](#) has the same structure: a quadratic form in the rescaled errors whose two factors are linear projections, so that the odds are an orthant probability. There the two random coordinates are the static shape error $\alpha - 1$ and the prior-variance error $\beta - 1$, and the quadratic form factorizes into the projections U_Λ and W_Λ . When one mode has fixed size and the other is random, the orthant probability is replaced by the product tail $T_\rho(z)$ of (2.19). This is the situation of [Section A](#) and [Section B](#).

6.3. Random localization exponents.

Proposition 6.4. Let $a = sZAP$ with a random exponent $P \in (0, 2)$ independent of a scalar Z , $\mathbb{P}(Z = 0) = 0$, and $\mathbb{P}(P = p_c(\gamma)) = 0$. Then

$$\lim_{\Lambda \rightarrow \infty} \lim_{s \downarrow 0} \mathbb{P}(h(\gamma) < h(0)) = \mathbb{P}(P < p_c(\gamma)).$$

With both modes and independent amplitudes, the limit is the probability of the SDE region of [Fig. 2](#), namely $\mathbb{P}((P_a, P_u, Z_u/Z_a) \in \mathcal{S}_\gamma)$ with $\mathcal{S}_\gamma := \{\frac{1}{4}\Delta L_\gamma(p_a) + \frac{1}{2}\rho^2 \Delta M_\gamma(p_u) < 0\}$.

Proof. Condition on P . Apply [Lemma 6.1](#), then [Theorem 5.3](#), which gives pointwise convergence of $\mathbf{1}\{L_\gamma < L_0\}$ for $p \neq p_c(\gamma)$, and conclude by dominated convergence. $\square$

This formulation turns the odds question into a question about the distribution of localization exponents across trained models. For the exponential family the corresponding limit is $\mathbb{P}(\kappa < 1)$ for every $\gamma > 0$ ([Proposition 5.6](#)).

7. THE ODDS II: GAUSSIAN PROCESSES WITH POWER-LAW ENVELOPES

7.1. Model. The measured profiles in [Section 9](#) change sign along the trajectory, share a systematic component across checkpoints, and have a root-mean-square magnitude that follows a piecewise power law. We therefore model

$$(7.1) \quad a(\sigma) = s[m_a(\sigma) + A(\sigma)G_a(x)], \quad u(\sigma) = s[m_u(\sigma) + U(\sigma)G_u(x)], \quad x := \log(\sigma/\sigma_0),$$

where A, U are deterministic envelopes (for example the power laws (5.1)), m_a, m_u are deterministic mean profiles, and G_a, G_u are centered, unit-variance, stationary Gaussian processes. We write their correlation functions as $k_a(\cdot/\ell_a)$ and $k_u(\cdot/\ell_u)$, with $k(0) = 1$ and correlation lengths ℓ_a, ℓ_u ; the two processes may be cross-correlated. Stationarity in $\log \sigma$ is a modelling choice. It is natural because networks trained with the EDM preconditioning receive the noise level through $c_{\text{noise}} = \frac{1}{4} \log \sigma$ [3]. The ensemble root mean square is $R_a^2 = m_a^2 + A^2$, so a fitted RMS curve estimates the envelope only when the ensemble mean is negligible [5].

In the variable x , $dy = 2y dx$ and the response functionals are integrals against the weights

$$(7.2) \quad w_\gamma(x) := 2(1 + \gamma) y(1 + y)^{-1-\gamma}, \quad w_\gamma^u(x) := (1 + \gamma) y(1 + y)^{-\frac{3+\gamma}{2}}, \quad y = e^{2x}.$$

The vector $\Xi := (L_0[a], L_\gamma[a], M_0[u], M_\gamma[u])/s$ is Gaussian, with means $L_\gamma[m_a]$, $M_\gamma[m_u]$ and, for instance,

$$(7.3) \quad C_{\gamma\eta}^a := \text{Cov}(L_\gamma[a], L_\eta[a])/s^2 = \iint w_\gamma(x) A(x) k_a\left(\frac{x-x'}{\ell_a}\right) A(x') w_\eta(x') dx dx'.$$

The infinite-dimensional prior thus enters the leading-order odds only through a 4-dimensional mean vector and a 4×4 covariance matrix.

7.2. Shape-only odds.

Theorem 7.1 (Orthant formula). *Assume exact mean and prior. Let $Z_0 := L_0[a]/s$ and $Z_\gamma := L_\gamma[a]/s$, with means μ_0, μ_γ and covariances $V_0 := C_{00}^a$, $V_\gamma := C_{\gamma\gamma}^a$, $C_{0\gamma} := C_{0\gamma}^a$, and assume that $Z_0 \pm Z_\gamma$ are nondegenerate. Put*

$$\rho_\gamma := \frac{V_0 - V_\gamma}{\sqrt{(V_0 + V_\gamma - 2C_{0\gamma})(V_0 + V_\gamma + 2C_{0\gamma})}}, \quad \nu_\mp := \frac{\mu_0 \mp \mu_\gamma}{\sqrt{V_0 + V_\gamma \mp 2C_{0\gamma}}}.$$

Then, as $s \downarrow 0$,

$$(7.4) \quad \mathbb{P}(h(\gamma) < h(0)) \rightarrow \Phi_2(\nu_-, \nu_+; \rho_\gamma) + \Phi_2(-\nu_-, -\nu_+; \rho_\gamma).$$

In the centered case $m_a \equiv 0$ this is $\frac{1}{2} + \frac{1}{\pi} \arcsin \rho_\gamma$, which exceeds $1/2$ if and only if $\text{Var } L_\gamma[a] < \text{Var } L_0[a]$.

Proof. By (4.1), the event is $Z_\gamma^2 < Z_0^2$ up to a set of vanishing probability, that is, $(Z_0 - Z_\gamma)(Z_0 + Z_\gamma) > 0$. The pair $(Z_0 - Z_\gamma, Z_0 + Z_\gamma)$ is bivariate normal with the stated standardized means and correlation. Split according to the common sign and use (2.18) in the centered case. $\square$

7.3. Two limits of the correlation length.

Theorem 7.2 (Sharp transition versus smooth odds). *Assume $m_a \equiv 0$ and that $w_\gamma A \in L^1 \cap L^2(dx)$ for the values of γ considered.*

- (i) (Long correlation.) *If k_a is continuous at 0, then as $\ell_a \rightarrow \infty$ the odds tend to $\mathbf{1}\{|L_\gamma[A]| < |L_0[A]|\}$ whenever $|L_\gamma[A]| \neq |L_0[A]|$: the process becomes a random scalar times A , and Lemma 6.1 applies.*
- (ii) (Short correlation.) *If $k_a \in L^1$ with $\kappa_1 := \int k_a \neq 0$, then $C_{\gamma\eta}^a/\ell_a \rightarrow \kappa_1 \hat{C}_{\gamma\eta}$ with $\hat{C}_{\gamma\eta} := \int w_\gamma w_\eta A^2 dx$, and the odds tend to $\frac{1}{2} + \frac{1}{\pi} \arcsin \hat{\rho}_\gamma$, where $\hat{\rho}_\gamma$ is ρ_γ computed from $\hat{C}$. For $A = A_p$ with $0 < p < 2$ and $\Lambda \rightarrow \infty$,*

$$(7.5) \quad \hat{C}_{\gamma\eta} = 2(1 + \gamma)(1 + \eta) B(2 - p, p + \gamma + \eta).$$

Consequently the white-noise odds exceed $1/2$ if and only if $(1 + \gamma)^2 B(2 - p, p + 2\gamma) < B(2 - p, p)$, that is, $p < p_{\text{wn}}(\gamma)$. In particular $p_{\text{wn}}(1) = (\sqrt{7} - 1)/2 \approx 0.823$, and $p_{\text{wn}}(\gamma) \rightarrow 1$ as $\gamma \downarrow 0$.

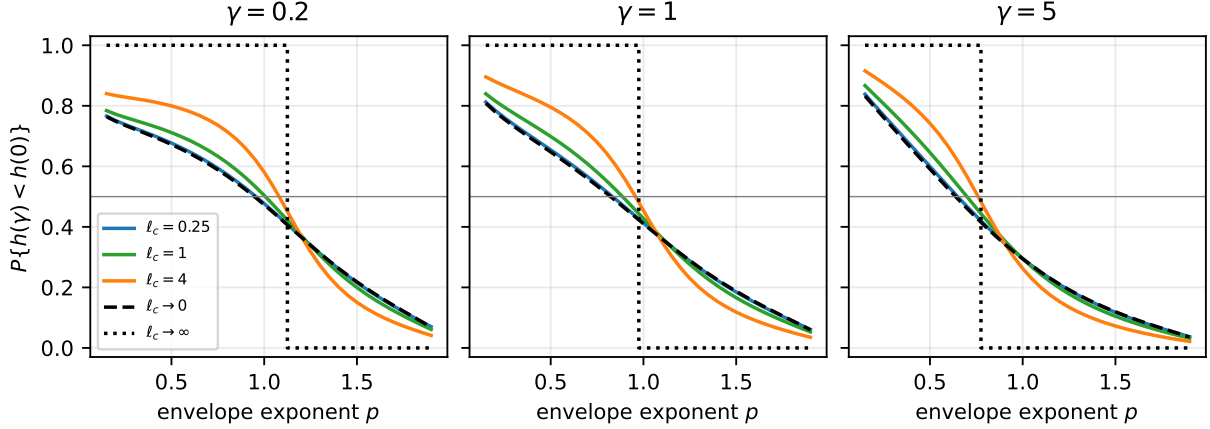


FIGURE 3. Leading-order probability that stochasticity helps, for centered Gaussian-process shape profiles $a = s A_p(\sigma) G(\log \sigma / \sigma_0)$ with exponential correlation $e^{-|\Delta x|/\ell_c}$, as a function of the envelope exponent p ($\Lambda \rightarrow \infty$). Dotted: $\ell_c \rightarrow \infty$ (step at $p_c(\gamma)$). Dashed: $\ell_c \rightarrow 0$, formula (7.5).

Proof. (i) $k_a((x - x')/\ell_a) \rightarrow 1$ pointwise, and dominated convergence (with $w_\gamma A \in L^1$) gives convergence of the covariance matrix to the rank-one matrix $(L_\gamma[A] L_\eta[A])_{\gamma\eta}$. Then $\rho_\gamma \rightarrow \text{sign}(L_0^2 - L_\gamma^2)$. (ii) $\ell_a^{-1} k_a(\cdot/\ell_a)$ is an approximate identity of mass κ_1 , and $w_\gamma A \in L^2$. The correlation ρ_γ is invariant under the common factor $\ell_a \kappa_1$. For (7.5), $w_\gamma w_\eta A_p^2 dx = 2(1 + \gamma)(1 + \eta)y^{1-p}(1 + y)^{-2-\gamma-\eta} dy$. For the small- γ limit, $\partial_\gamma \log[(1 + \gamma)^2 B(2 - p, p + 2\gamma)]|_{\gamma=0} = 2(1 + \psi(p) - \psi(2)) = 2(\psi(p) - \psi(1))$, which vanishes at $p = 1$. For $\gamma = 1$ the condition reads $4\Gamma(p + 2)/\Gamma(4) < \Gamma(p)$, that is, $2p^2 + 2p - 3 < 0$. $\square$

The correlation length therefore interpolates between the sharp phase transition of Section 5 ($\ell_a = \infty$) and an explicit Beta-function odds curve ($\ell_a \rightarrow 0$). Figure 3 shows intermediate values for an exponential (Ornstein–Uhlenbeck) correlation in $\log \sigma$. For $\gamma = 1$, the white-noise odds are 0.78, 0.65, 0.51, 0.41, 0.27 and 0.12 at $p = 0.2, 0.5, 0.8, 1.0, 1.3$ and 1.7 . Temporal decorrelation of the score error smears the transition into odds.

7.4. Both modes: quadratic forms. With both modes present, the event $h(\gamma) < h(0)$ is, at leading order, $\{\Xi^\top K \Xi < 0\}$ with $K = \text{diag}(-\frac{1}{4}, \frac{1}{4}, -\frac{1}{2}, \frac{1}{2})$, an indefinite quadratic form in a 4-dimensional Gaussian vector; in general no arcsine reduction exists. In the centered case, let $\lambda_1, \dots, \lambda_4$ be the eigenvalues of $\Sigma^{1/2} K \Sigma^{1/2}$, where $\Sigma = \text{Cov}(\Xi)$. Imhof's formula [2] gives

$$(7.6) \quad \mathbb{P}(h(\gamma) < h(0)) \rightarrow \frac{1}{2} - \frac{1}{\pi} \int_0^\infty \frac{\sin(\frac{1}{2} \sum_j \arctan(\lambda_j t))}{t \prod_j (1 + \lambda_j^2 t^2)^{1/4}} dt,$$

with the noncentral version available when $m_a, m_u \neq 0$. When the two modes are independent the eigenvalues split into two pairs of opposite signs, one per mode.

7.5. Nonstationary examples and finite bases.

Example 7.3 (Brownian profile). Let $a = s B_\ell$ with B a Brownian motion in the sampling clock, started at the beginning of sampling and independent of the sampler, with exact mean and prior. The local comparison has the sign of $L_0 \partial_\gamma L_0 = MN$, with $M = \int_0^\Lambda B_\ell d\ell$ and $N = \int_0^\Lambda (\ell + 1 - \Lambda) B_\ell d\ell$. Since $\text{Cov}(M, N) = \Lambda^3(8 - 3\Lambda)/24$,

$$\lim_{s \downarrow 0} \mathbb{P}(h'(0) < 0) \geq \frac{1}{2} \iff \Lambda \geq \frac{8}{3}.$$

The threshold $\Lambda = 2$ of [Proposition 3.4](#) therefore belongs to constant profiles and is not universal.

Example 7.4 (Positive diffusion profiles). Let $A(\ell) = A_0 \exp\{\nu B_\ell - \nu^2 \ell/2\}$ with $A_0 > 0$ of any law. Then A_0 cancels from the centroid of [Theorem 4.2](#), and as $\nu \downarrow 0$, $\bar{d}_A = \Lambda/2 + \nu Z_\Lambda + o_P(\nu)$ with $Z_\Lambda \sim \mathcal{N}(0, \Lambda^3/120)$. The local shape-only odds therefore tend to 1, 0 or 1/2 according as $\Lambda > 2$, < 2 or $= 2$. At positive volatility the answer is determined by the distribution of the random centroid.

Remark 7.5 (Finite bases and estimation). Replacing G by $\sum_{j=1}^m \xi_j \phi_j$ with $\xi \sim \mathcal{N}(0, \Sigma)$ gives $\text{Cov}(L_\gamma, L_\eta) = k_\gamma^\top \Sigma k_\eta$ with $(k_\gamma)_j = L_\gamma[A\phi_j]$, a practical route to fitting priors on measured profiles. Given an ensemble of checkpoints, m and R are estimated by the pointwise mean and root mean square, $A = \sqrt{R^2 - m^2}$. The correlation length can be read from the empirical autocorrelation of $(a-m)/A$ in $\log \sigma$, or, for a differentiable kernel, from the rate of zero crossings via Rice's formula [\[6\]](#): $\mathbb{E}[\#\text{crossings per unit } x] = \sqrt{-k''(0)}/(\pi\ell)$.

8. BEYOND THE PERTURBATIVE REGIME

The formulas of [Proposition 2.1](#) are exact for arbitrary profiles, so some statements survive without any smallness assumption. We record the local ($\gamma \downarrow 0$) versions for a positive shape path $\alpha = (\alpha(\ell))_{0 \leq \ell \leq \Lambda}$ with $\int_0^\Lambda \alpha^{-1} < \infty$, a scalar prior shape $\beta_T > 0$ that may depend on the path, and static Gaussian mean errors $X = \mu_\theta - \mu_0 \sim \mathcal{N}(0, s_\theta^2)$ and $Y = \mu_T - \mu_0 \sim \mathcal{N}(0, s_T^2)$. The mean errors are independent of each other and of (α, β_T) , and everything is independent of the sampler noise. Define

(8.1)

$$I_\alpha(s) := \int_s^\Lambda \frac{dr}{\alpha(r)}, \quad I_\alpha := I_\alpha(0), \quad E_\alpha := e^{\Lambda - I_\alpha}, \quad R_\alpha := e^{-I_\alpha/2}, \quad J_\alpha := \int_0^\Lambda e^{(\Lambda-s)-I_\alpha(s)} ds.$$

In the coordinates [\(2.6\)](#), $I_\alpha(s) = (\Lambda - s) - I_a(s)$, $E_\alpha = e^{I_a(0)}$ and $J_\alpha = \int_0^\Lambda e^{I_a(s)} ds$.

Proposition 8.1 (Exact terminal moments). *Conditionally on (α, β_T, X, Y) ,*

$$r(\gamma) = \frac{X + (Y - X)e^{-(1+\gamma)I_\alpha/2}}{\sigma_0}, \quad v(\gamma) = \beta_T e^{\Lambda - (1+\gamma)I_\alpha} + \gamma \int_0^\Lambda e^{(\Lambda-s) - (1+\gamma)I_\alpha(s)} ds.$$

At $\gamma = 0$: $v = \beta_T E_\alpha$ and $\dot{v} := v'(0) = J_\alpha - I_\alpha \beta_T E_\alpha$.

Put $S := \dot{v}(v-1)/(2v)$, $G_\alpha := (1 - R_\alpha)X + R_\alpha Y$, $H := Y - X$, $\kappa_\alpha := I_\alpha R_\alpha/2$, $\tau_\alpha := \kappa_\alpha/\sigma_0^2$, $\delta_{\alpha,\beta} := \dot{v}/(2v\kappa_\alpha)$ and $C_{\alpha,\beta} := H + \delta_{\alpha,\beta} G_\alpha$. Then, exactly,

(8.2)

$$h'_q(0) = S - \tau_\alpha G_\alpha H, \quad h'_c(0) = \frac{1}{v}(S - \tau_\alpha G_\alpha C_{\alpha,\beta}), \quad S = -\frac{I_\alpha E_\alpha}{2\beta_T}(\beta_T - B_0[\alpha])(\beta_T - B_1[\alpha]),$$

with the path-dependent band $B_0[\alpha] := E_\alpha^{-1}$ and $B_1[\alpha] := J_\alpha/(I_\alpha E_\alpha)$. For a constant path these reduce to [\(3.3\)](#).

Theorem 8.2 (Exact mixture formula). *Let $V_G := (1 - R_\alpha)^2 s_\theta^2 + R_\alpha^2 s_T^2$, $V_H := s_\theta^2 + s_T^2$, $K_{GH} := R_\alpha s_T^2 - (1 - R_\alpha) s_\theta^2$ and $V_C := V_H + 2\delta K_{GH} + \delta^2 V_G$. Define*

$$\rho_q := \frac{K_{GH}}{\sqrt{V_G V_H}}, \quad z_q := \frac{S}{\tau_\alpha \sqrt{V_G V_H}}, \quad \rho_c := \frac{K_{GH} + \delta V_G}{\sqrt{V_G V_C}}, \quad z_c := \frac{S}{\tau_\alpha \sqrt{V_G V_C}}.$$

If $s_\theta, s_T > 0$, then for any joint law of (α, β_T) ,

$$\mathbb{P}(h'_q(0) < 0) = \mathbb{E}[T_{\rho_q}(z_q)], \quad \mathbb{P}(h'_c(0) < 0) = \mathbb{E}[T_{\rho_c}(z_c)].$$

Consequently $\mathbb{E}[\mathbf{1}_{\{S \leq 0\}} q(\rho_j)] \leq \mathbb{P}(h'_j(0) < 0) \leq \mathbb{P}(S < 0) + \mathbb{E}[\mathbf{1}_{\{S \geq 0\}} q(\rho_j)]$ with $q(\rho) := \frac{1}{2} + \frac{1}{\pi} \arcsin \rho$, and $\rho_q \geq 0$ if and only if $I_\alpha \leq 2 \log(1 + s_T^2/s_\theta^2)$.

Proof. Conditionally on the path, G_α , H and $C_{\alpha,\beta}$ are centered Gaussian linear forms in (X, Y) , and both conditional covariance determinants equal $s_\theta^2 s_T^2 > 0$. By (8.2) the events are $G_\alpha H > S/\tau_\alpha$ and $G_\alpha C_{\alpha,\beta} > S/\tau_\alpha$. Standardize, apply (2.19), and take expectations. The bounds use monotonicity of T_ρ . $\square$

Proposition 8.3 (Exact shape-only odds). *If $X = Y = 0$, then for both orientations $\mathbb{P}(h'(0) < 0) = 1 - \mathbb{E}[\Pi_\alpha]$, where $\Pi_\alpha := \mathbb{P}(B_-[\alpha] \leq \beta_T \leq B_+[\alpha] \mid \alpha)$. If β_T is independent of the path with continuous distribution function F_β , then $\Pi_\alpha = |F_\beta(B_1[\alpha]) - F_\beta(B_0[\alpha])|$. Without further assumptions on the path law, any value in $[0, 1]$ can occur.*

Remark 8.4 (Open: finite-amplitude power laws). For the measured profiles the full Gaussian moment dynamics and the first-order response agree closely (Section 9), so the perturbative regime appears to be the relevant one in practice. The leading-order comparison is, however, invariant under $a \mapsto -a$, $u \mapsto -u$. The first correction, of order ε^3 , distinguishes over-dispersed ($\alpha > 1$) from under-dispersed ($\alpha < 1$) scores. Near $p = p_c(\gamma)$ it should smooth the sharp transition of Theorem 5.3 into one of width proportional to the amplitude, even for a scalar random amplitude. Computing this correction requires the second variation of (2.9), which is explicit but not carried out here.

9. COMPARISON WITH TRAINED NETWORKS ON A GAUSSIAN MIXTURE

[Note: All numbers in this section come from notebook 056-stoch_tests-wo_fits. The envelope fits were made on $(\log \alpha_\theta, (\mu_\theta - \mu_0)/\sqrt{V_{\text{ref}}})$ and are to be redone on the linear coordinates (a, u) , which are defined at every grid point and avoid the singularity at $a = 1$. Figure 4 is computed from the provisional fits; figures of the raw profiles are placeholders.]

9.1. Setup. We use the one-dimensional Gaussian mixture of [7] [TODO: component weights, means and standard deviations], with mean $\mu_0 = -0.01$ and variance $\sigma_0^2 = 0.1219$, so $V(t) = 0.1219 + t^2$ under the EDM schedule on $\sigma \in [0.002, 80]$; the horizon is $\Lambda = \log(V(80)/V(0.002)) \approx 10.87$. There are 144 checkpoints from three training runs (epochs 0–47) [TODO: architecture and training details]. The checkpoints are not independent draws: epochs within a run are strongly correlated. Sampling uses 500 Euler (ODE) or Euler–Maruyama (SDE) steps.

9.2. Projected score-error profiles. At each noise level we project the network error $\epsilon_\theta = s_\theta - s_{\text{exact}}$ onto affine functions in $L^2(p_t)$,

$$\epsilon_\theta(x, t) = b(t) + C(t)(x - \mu_0) + \varrho(x, t), \quad \mathbb{E}_{p_t}[\varrho] = \mathbb{E}_{p_t}[(X - \mu_0)\varrho] = 0,$$

using 256 Gauss–Hermite nodes per mixture component. The mean, affine and residual energies then add up to $\mathbb{E}_{p_t}[\epsilon_\theta^2]$. The linear coordinates of (2.6) are $a = VC$ and $u = Vb/\sigma_0$; the notebook uses $\sqrt{V_{\text{ref}}} = \sqrt{V(0.002)}$ in place of σ_0 , a relative difference of 2×10^{-5} . Among the 84 checkpoints with epoch ≥ 20 , 329 of 24,192 grid points, in 27 checkpoints, have $a \geq 1$ and hence no positive α_θ . The linear coordinates are unaffected. [TODO: Figure: signed profiles $a(\sigma)$ and $u(\sigma)$ for the epoch- ≥ 20 cohort, colored by epoch, with the RMS envelopes; histogram of the integrated residual-energy fraction.]

9.3. Power-law envelopes. Table 3 summarizes continuous piecewise power-law fits to the ensemble root mean square, obtained by weighted least squares in $\log R$ with uniform weight in $\log \sigma$. Three observations matter for the theory.

- (a) At low noise the shape envelope decays like $\sigma^{-0.7}$, below $p_c(\gamma)$ for all $\gamma \lesssim 10$. It steepens to $\sigma^{-1.3}$ – $\sigma^{-1.5}$ above $\sigma \approx 0.5$ – 0.9 , that is, 1.5 – $2.5 \sigma_0$.
- (b) The mean envelope grows toward the data end like $\sigma^{-0.9}$ to $\sigma^{-1.2}$ and is flat above $\sigma \approx 0.36$ – $0.39 \approx \sigma_0$. By Theorem 5.9, the mean mode is ODE-favorable at every γ .

coord.	cohort	K	exponents (low $\rightarrow$ high σ)	breaks in σ	$\log_{10}\text{RMSE}$	R^2	mean fr.
$\log \alpha_\theta$	each t	2	0.695, 1.331	0.524	0.071	0.948	0.32
$\log \alpha_\theta$	each t	3	0.711, 1.504, 0.875	0.856, 17.8	0.054	0.935	0.32
$\log \alpha_\theta$	throughout	3	0.701, 1.504, 0.876	0.851, 17.1	0.042	0.953	0.68
u	throughout	2	0.924, 0	0.364	0.027	0.976	0.26
u	throughout	3	1.198, 0.886, 0	0.0062, 0.391	0.018	0.999	0.26

TABLE 3. Piecewise power-law fits $R \propto \sigma^{-p_j}$ to the RMS across the 84 checkpoints with epoch ≥ 20 (“throughout”: the 57 checkpoints with $a < 1$ at every noise level). The last column is the fraction of the weighted mean square carried by the ensemble mean. Provisional: fitted on $\log \alpha_\theta$ and $(\mu_\theta - \mu_0)/\sqrt{V_{\text{ref}}}$.

γ	L_γ/L_0 (shape)	M_γ/M_0 (mean)	$\log_{10} \mathbb{E}h(\gamma)/\mathbb{E}h(0)$	two-amplitude odds
0.2	0.95–0.96	1.05	−0.005–0.000	0.50–0.54
1	0.90–0.94	1.25	+0.05–+0.06	0.33–0.39
5	1.06–1.11	1.99–2.00	+0.36–+0.37	0

TABLE 4. Leading-order predictions from the RMS envelopes of Table 3 (ranges over the 2- and 3-exponent fits), with the exact finite horizon $\Lambda = 10.87$ and $\sigma_0^2 = 0.1219$.

- (c) The ensemble mean is not negligible for the shape mode. It carries 32% of the weighted mean square for all checkpoints and 68% for the cohort valid throughout, against 2–26% for the mean mode. The Gaussian-process model (7.1) therefore needs $m_a \neq 0$, and the relevant odds are the noncentered orthant probabilities of Theorem 7.1.

The breaks sit near the data scale, which suggests the prediction that they scale with σ_0 when the mixture parameters are varied. The break of the mean envelope at $\sigma \approx 0.006$ may instead track the component scale. **[TODO: test with randomized mixtures.]**

9.4. Predictions from the envelopes. Applying the finite-horizon kernels (2.13)–(2.14) to the fitted envelopes gives Table 4 and Fig. 4. Here the envelopes are treated as deterministic profiles with independent random signs, which ignores the systematic component noted in (c). The shape mode is mildly SDE-favorable, with $\gamma^* \approx 0.8$ –1.0 and $\gamma_c \approx 2.6$ –3.7. The largest reduction of L_γ is only 6–10%, because the steeper segment above $1.5\sigma_0$ removes the mass that stochasticity would damp. For comparison, a pure $\sigma^{-0.7}$ profile gives $L_1/L_0 = 0.71$. The mean mode is ODE-favorable and dominates for $\gamma \gtrsim 0.5$. The predicted KL ratio is therefore flat at small γ and increases for $\gamma \gtrsim 1$. The two-amplitude odds of Theorem 6.2 with unit scales are 0.50–0.54 at $\gamma = 0.2$, 0.33–0.39 at $\gamma = 1$ and 0 at $\gamma = 5$. The $L^1(d\ell)$ norm of the shape envelope is 0.17 (Remark 2.4).

9.5. Controlled residual ablation. To separate the affine part of the error from the nonlinear residual, the notebook compares the samplers with scores $s_\lambda = s_{\text{exact}} + b + C(x - \mu_0) + \lambda g$ for $\lambda = 1$ (full learned score) and $\lambda = 0$ (affine error only). The comparison uses an exact mixture prior, matched initial states and Brownian increments, and $\gamma \in \{0, 0.01, 0.2, 1, 5\}$, with a measured ODE baseline. KL divergences are estimated from 4 seeds of 4096 particles in 64 target-quantile bins. For each checkpoint the *profile prediction* $\log_{10}[\hat{h}(\gamma)/\hat{h}(0)]$ uses (2.17) with the measured signed profiles (a, u) , and the *moment prediction* integrates (2.8)–(2.9) exactly (500 Euler steps) and evaluates (2.12). Table 5 summarizes the agreement.

The following conclusions are provisional.

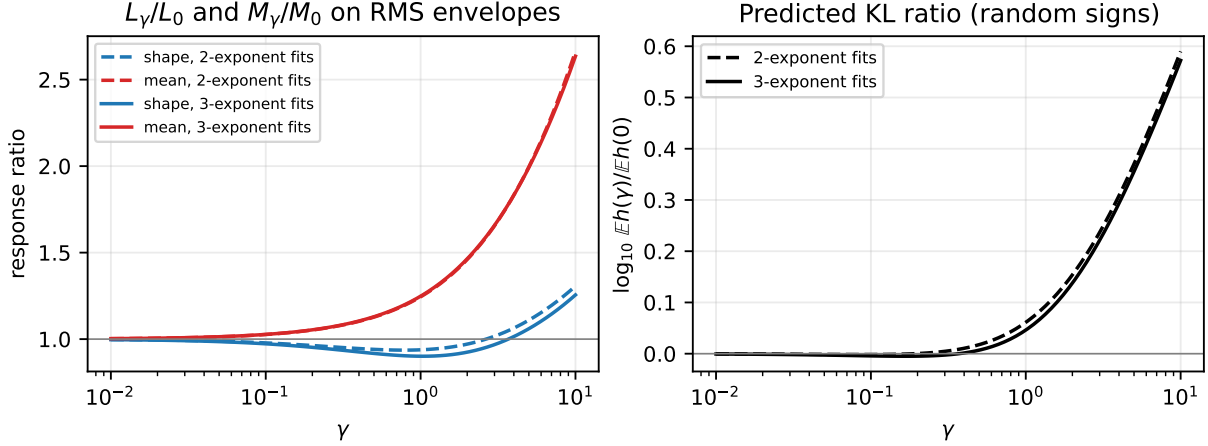


FIGURE 4. Response ratios and predicted KL ratio as functions of γ , computed from the RMS envelopes of Table 3.

γ	arm	Spearman ($q\ p$ / $p\ q$)	log-bias, dex ($q\ p$ / $p\ q$)	moments vs. profile
0.2	full	0.15 / 0.13	+0.14 / +0.15	identical to ± 0.03
	affine	0.50 / 0.51	−0.04 / −0.04	
1	full	0.43 / 0.39	+0.30 / +0.29	
	affine	0.57 / 0.57	+0.02 / +0.01	
5	full	0.52 / 0.50	+0.52 / +0.42	
	affine	0.60 / 0.60	+0.19 / +0.17	

TABLE 5. Agreement between the first-order profile prediction and the measured $\log_{10}[\text{KL}(\gamma)/\text{KL}(0)]$ over 144 checkpoints. Log-bias is the mean of prediction minus measurement. The last column compares with the full Gaussian moment dynamics.

- (i) The linearization of Lemma 2.2 is not the limiting approximation: profile and moment predictions are nearly indistinguishable.
- (ii) For the affine-only sampler the Gaussian theory is nearly unbiased at $\gamma \leq 1$ and has moderate rank agreement.
- (iii) The full score is shifted toward stochasticity relative to the prediction, by about 0.3 dex at $\gamma = 1$ and 0.4–0.5 dex at $\gamma = 5$. Removing the residual reduces the prediction gap in 67–68% of checkpoints at $\gamma = 1$, with median gap going from 0.29–0.30 to 0.19–0.20 dex. The nonlinear residual is thus systematically SDE-favorable, and this is not captured by the affine theory.
- (iv) Log-ratio metrics are ill-conditioned when the signed baseline functional $L_0[a]$ nearly cancels. This explains negative skill scores and very large ratio-RMSE despite reasonable rank agreement. **[TODO: report sign agreement (confusion matrix) and the bounded statistic $(h_\gamma - h_0)/(h_\gamma + h_0)$.]**

9.6. Empirical odds. In the original sweeps, whose smallest recorded $\gamma_0 \approx 0.01$ serves as the baseline, the statistic $I = \text{KL}(\gamma_0)/\min_{j \geq 3} \text{KL}(\gamma_j)$ exceeds 1.01 for 91.0% ($q\|p$) and 92.4% ($p\|q$) of checkpoints. This “some γ helps” event is not the fixed- γ event of the theorems; it contains $\{h'(0) < 0\}$ and is selected in-sample. **[TODO: fixed- γ win rates from the ablation (both arms),**

to compare with Table 4, Theorem 6.2 and Theorem 7.1 with fitted (m_a, A, ℓ_a) ; estimate ℓ_a from the autocorrelation of $(a - m_a)/A$ in $\log \sigma$.]

10. DISCUSSION AND OPEN PROBLEMS

The time-dependent formulation changes what the Gaussian model is about. The central object is no longer a collection of static misspecification parameters but an error profile along the trajectory, and the relevant statistics are a few linear projections of that profile. Four conclusions stand out.

First, arbitrary time dependence is analytically natural. In the coordinates (a, u) the reverse moments obey linear equations, and the second-order KL is a sum of squares of two exponential filters of the profile, with a remainder controlled in $L^1(d\ell)$.

Second, stochasticity has a sharp physical interpretation: it amplifies shape error below $\sigma_a^*(\gamma)$ and damps it above, and $\sigma_a^*(1) = \sigma_0$.

Third, for power-law profiles, which describe trained networks well in magnitude, the critical shape exponent depends on γ and satisfies $p_c(1) = 1$. The mean threshold stays at $p_u = 0$. Optimal and maximal useful stochasticities follow from digamma identities.

Fourth, the odds reduce to orthant probabilities or quadratic forms in low-dimensional Gaussian vectors. For Gaussian processes with power-law envelopes, the correlation length controls whether the ODE–SDE comparison behaves as a phase transition or as a smooth odds curve.

On the Gaussian mixture, the theory accounts for the affine part of the error, including the shape of the γ -dependence and the penalty at large γ driven by a mean error that grows toward the data end, but not for the additional benefit of stochasticity carried by the nonlinear residual. The following problems appear most pressing.

- (i) *The nonlinear residual.* Explain, for example through the KL-evolution identities of [7], why the residual that is orthogonal to affine functions systematically favors stochasticity.
- (ii) *Finite amplitude.* Compute the $O(\varepsilon^3)$ correction and determine whether the sharp transition at $p_c(\gamma)$ persists or broadens.
- (iii) *Optimal schedules.* Minimize $|L_{\gamma(\cdot)}[a]|$ and $|M_{\gamma(\cdot)}[u]|$ in (4.6), or the leading-order KL, over schedules $\gamma(\ell)$ for power-law profiles. Proposition 5.5 is the constant-schedule case.
- (iv) *Higher dimensions.* For $s_\theta(x, t) = -A_\theta(t)(x - \mu_\theta(t))$ the covariance obeys a matrix Lyapunov equation; determine whether each eigendirection of the score Jacobian carries its own critical exponent.
- (v) *Identifiability.* Determine which training-time diagnostics estimate L_γ and M_γ without access to the exact score.
- (vi) *Universality of the envelope.* Test whether the exponents and breaks of the envelope are reproducible across datasets and scale with the data and component scales.
- (vii) *Proof of ?? 5.4.*

APPENDIX A. THE STATIC BENCHMARK: RANDOM MEANS AND GAMMA PRIORS

This appendix condenses Sections 2.1 and 3 of the static draft. The notation is that of Section 3.

A.1. Random Gaussian means under general shape priors. Let $X = \mu_\theta - \mu_0 \sim \mathcal{N}(0, s_\theta^2)$ and $Y = \mu_T - \mu_0 \sim \mathcal{N}(0, s_T^2)$ be independent of each other and of the positive shape vector (α, β) , which need not have independent components. Put

$$A_\alpha := (1 - R_\alpha)X + R_\alpha Y, \quad B := Y - X, \quad \kappa_\alpha := \frac{R_\alpha \Lambda}{2\alpha}, \quad \tau_\alpha := \frac{\kappa_\alpha}{\sigma_0^2},$$

$$\delta_{\alpha,\beta} := \frac{v'(0)}{2v(0)\kappa_\alpha}, \quad C_{\alpha,\beta} := B + \delta_{\alpha,\beta}A_\alpha.$$

Then, exactly, $h'_q(0) = S - \tau_\alpha A_\alpha B$ and $h'_c(0) = v(0)^{-1}(S - \tau_\alpha A_\alpha C_{\alpha,\beta})$. With $V_A := (1 - R_\alpha)^2 s_\theta^2 + R_\alpha^2 s_T^2$, $V_B := s_\theta^2 + s_T^2$, $K_{AB} := R_\alpha s_T^2 - (1 - R_\alpha)s_\theta^2$, $V_C := V_B + 2\delta K_{AB} + \delta^2 V_A$, and $(\rho_q, z_q, \rho_c, z_c)$ defined as in [Theorem 8.2](#) (with $G \rightarrow A$, $H \rightarrow B$):

Proposition A.1 (Exact mixture formula). *For any joint law of (α, β) , $\mathbb{P}(h'_q(0) < 0) = \mathbb{E}[T_{\rho_q}(z_q)]$ and $\mathbb{P}(h'_c(0) < 0) = \mathbb{E}[T_{\rho_c}(z_c)]$. In particular $\mathbb{P}(h'_j(0) < 0) \geq \mathbb{E}[\mathbf{1}_{\{S \leq 0\}}(\frac{1}{2} + \frac{1}{\pi} \arcsin \rho_j)]$, and $\rho_q \geq 0$ if and only if $R_\alpha \geq s_\theta^2/(s_\theta^2 + s_T^2)$.*

Proposition A.2 (Limits). (i) (Small mean uncertainty.) *If (X, Y) is replaced by $(\varepsilon X, \varepsilon Y)$ and $\mathbb{P}(S = 0) = 0$, both probabilities tend to $\mathbb{P}(S < 0)$ as $\varepsilon \downarrow 0$.*
(ii) (Concentrated shapes.) *If $(\alpha_\varepsilon, \beta_\varepsilon) \rightarrow (1, 1)$ in probability, both probabilities tend to $\frac{1}{2} + \frac{1}{\pi} \arcsin \rho_{\text{mean}}$ with $r = \sqrt{\zeta_T}$ and*

$$\rho_{\text{mean}} = \frac{r s_T^2 - (1 - r)s_\theta^2}{\sqrt{((1 - r)^2 s_\theta^2 + r^2 s_T^2)(s_\theta^2 + s_T^2)}},$$

which exceeds 1/2 if and only if $\sqrt{\zeta_T} > s_\theta^2/(s_\theta^2 + s_T^2)$.
(iii) (Joint small error.) *If $((\alpha_\varepsilon - 1)/\varepsilon, (\beta_\varepsilon - 1)/\varepsilon) \Rightarrow (U, V)$ and the means are $(\varepsilon X, \varepsilon Y)$, both derivatives divided by ε^2 converge in law to*

$$Q_{\text{full}} = \frac{\Lambda}{4}(\Lambda U + V)((2 - \Lambda)U - 2V) - \frac{\Lambda\sqrt{\zeta_T}}{2\sigma_0^2}((1 - \sqrt{\zeta_T})X + \sqrt{\zeta_T}Y)(Y - X),$$

a four-dimensional quadratic form (for Gaussian (U, V)), for which Imhof's formula (7.6) applies.

Proposition A.3 (Strong noising with random means). *If $\mathbb{P}(\alpha = 1) = 0$, then $\mathbb{P}(h'_j(0) < 0) \rightarrow 1$ as $\zeta_T \downarrow 0$ for $j \in \{q, c\}$.*

Proof. For $\alpha < 1$, $S \rightarrow -\infty$ because $v(0) \rightarrow 0$ and $v'(0) \rightarrow (1/\alpha - 1)^{-1} > 0$. For $\alpha > 1$, $S \sim -\beta\Lambda e^{\Lambda(1-1/\alpha)}/(2\alpha) \rightarrow -\infty$. The mean terms stay bounded for h_q . For h_c , the additional term is dominated in both cases. $\square$

Corollary A.4 (Noncommuting limits). *Let $(\alpha_\varepsilon, \beta_\varepsilon) \rightarrow (1, 1)$ in probability with $\mathbb{P}(\alpha_\varepsilon = 1) = 0$. Then*

$$\lim_{\varepsilon \downarrow 0} \lim_{\zeta_T \downarrow 0} \mathbb{P}(h'_j(0) < 0) = 1, \quad \lim_{\zeta_T \downarrow 0} \lim_{\varepsilon \downarrow 0} \mathbb{P}(h'_j(0) < 0) = q^* := \frac{1}{2} - \frac{1}{\pi} \arcsin \frac{s_\theta}{\sqrt{s_\theta^2 + s_T^2}},$$

and $q^ = 1/4$ when $s_\theta = s_T$.*

Remark A.5 (Consistency check). At $\alpha = \beta = 1$ with only μ_θ random, $h'(0) = \Lambda r(1 - r)X^2/(2\sigma_0^2) > 0$ almost surely, where $r = \sqrt{\zeta_T}$. With only μ_T random, $h'(0) = -\Lambda r^2 Y^2/(2\sigma_0^2) < 0$ almost surely. [Note: This corrects a sign in the one-mean statement of the current JMLR manuscript.]

A.2. Gamma shape priors. Let $\alpha \sim \text{Gamma}(k_\alpha, r_\alpha)$ and $\beta \sim \text{Gamma}(k_\beta, r_\beta)$ be independent, with distribution functions $F_{k,r}(x) = \gamma_{\text{inc}}(k, rx)/\Gamma(k)$. By [Lemma 3.2](#),

$$\mathbb{P}(S < 0) = 1 - \int_0^\infty |F_{k_\beta, r_\beta}(B_1(a)) - F_{k_\beta, r_\beta}(B_0(a))| f_{k_\alpha, r_\alpha}(a) da.$$

Unlike $\text{LogNormal}(0, \sigma^2)$, a Gamma law cannot be simultaneously centered in median and logarithmic mean. Four useful centerings are: unit median, $\text{Gamma}(k, q_k)$ with q_k the median of $\text{Gamma}(k, 1)$; unit mean, $\text{Gamma}(k, k)$; unit mode, $\text{Gamma}(k, k-1)$; and zero log-mean, $\text{Gamma}(k, e^{\psi(k)})$. In all four the rate is $k + O(1)$. Matching the logarithmic variance of a lognormal prior means choosing k from $\psi_1(k) = \sigma^2$. The unit-median convention preserves the exact e^{-2} threshold of [Proposition 3.4](#); the unit-mean convention does not.

Proposition A.6 (Small dispersion). *With any of the four centerings, if $k_\alpha, k_\beta \rightarrow \infty$ with $k_\alpha/k_\beta \rightarrow \eta \in (0, \infty)$, then*

$$\mathbb{P}(h'_q(0) < 0) = \mathbb{P}(h'_c(0) < 0) = \frac{1}{2} + \frac{1}{\pi} \arcsin \frac{\Lambda(\Lambda - 2) + 2\eta}{\sqrt{(\Lambda^2 + \eta)((2 - \Lambda)^2 + 4\eta)}} + o(1).$$

The limit exceeds $1/2$ for every Λ if $\eta > 1/2$. For $0 < \eta < 1/2$ it is below $1/2$ exactly for $\Lambda \in (1 - \sqrt{1 - 2\eta}, 1 + \sqrt{1 - 2\eta})$.

Proof. The Gamma normal limit gives $((\alpha - 1)/\varepsilon, (\beta - 1)/\varepsilon) \Rightarrow (Z_\alpha, \sqrt{\eta}Z_\beta)$ with $\varepsilon = k_\alpha^{-1/2}$; apply [Proposition 3.3](#) with $s_{11} = 1$, $s_{12} = 0$, $s_{22} = \eta$. $\square$

At finite shape the lognormal and Gamma families differ in their behavior near zero and in logarithmic skewness, which is 0 for the lognormal and $\psi_2(k)/\psi_1(k)^{3/2} = -k^{-1/2} + O(k^{-3/2})$ for the Gamma law. Matching median and logarithmic variance yields no stochastic ordering of the sign probabilities, so finite-dispersion comparisons must use the exact formulas. Small shapes ($k \leq 1$) put substantial mass near zero and are outside the perturbative regime.

APPENDIX B. LOCAL ODDS WITH TIME-DEPENDENT SHAPE AND STATIC GAUSSIAN MEANS

This appendix condenses Section 4 of the static draft, in the coordinates (2.6). Let $a = \varepsilon A$ with continuous paths, $b_T = \varepsilon Z_\beta$, and static means X, Y as in [Section A](#). The pair (A, Z_β) may be dependent but is independent of (X, Y) . Put

$$\Delta := Z_\beta + \int_0^\Lambda A dd, \quad \Theta := \int_0^\Lambda (1 - d)A dd - \Lambda Z_\beta,$$

$$r := e^{-\Lambda/2}, \quad G := (1 - r)X + rY, \quad H := Y - X,$$

so that $\Delta = L_0[A; Z_\beta]$ and $\Theta = \partial_\gamma L_\gamma[A; Z_\beta]|_{\gamma=0}$. Let $V_G := (1 - r)^2 s_\theta^2 + r^2 s_T^2$, $V_H := s_\theta^2 + s_T^2$, $K_{GH} := r s_T^2 - (1 - r) s_\theta^2$ and $\rho_m := K_{GH}/\sqrt{V_G V_H}$.

Proposition B.1 (Fixed-size means dominate). *If the mean errors are not scaled with ε , then for both orientations $\mathbb{P}(h'(0) < 0) \rightarrow \frac{1}{2} + \frac{1}{\pi} \arcsin \rho_m$, which is above, at, or below $1/2$ according as $\sqrt{\zeta_T} \gtrless s_\theta^2/(s_\theta^2 + s_T^2)$.*

Proposition B.2 (Joint scaling). *If the means are $(\varepsilon X, \varepsilon Y)$, then $\varepsilon^{-2} h'(0) \rightarrow Q_{\text{td}} := \frac{1}{2} \Delta \Theta - \frac{\Lambda r}{2\sigma_0^2} GH$ almost surely for both orientations, and*

$$\lim_{\varepsilon \downarrow 0} \mathbb{P}(h'(0) < 0) = \mathbb{E} \left[T_{\rho_m} \left(\frac{\sigma_0^2 \Delta \Theta}{\Lambda r \sqrt{V_G V_H}} \right) \right].$$

If $\rho_m \geq 0$ and $\Delta\Theta \leq 0$ almost surely, this is at least $1/2$. If $\rho_m \leq 0$ and $\Delta\Theta \geq 0$ almost surely, it is at most $1/2$. For $Z_\beta = 0$ and $A \geq 0$, $\Delta\Theta = (\int A)^2(1 - \bar{d}_A)$, which combines the centroid criterion of [Theorem 4.2](#) with the mean correlation ρ_m .

Proof. $v(0) - 1 = \varepsilon\Delta + O(\varepsilon^2)$ and $v'(0) = \varepsilon\Theta + O(\varepsilon^2)$ by [Lemma 2.2](#) and [\(4.5\)](#), so $S = \frac{1}{2}\varepsilon^2\Delta\Theta + O(\varepsilon^3)$. The mean terms follow from [\(3.2\)](#) with $R_\alpha = r + O(\varepsilon)$. Condition on (A, Z_β) and apply [\(2.19\)](#). $\square$

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